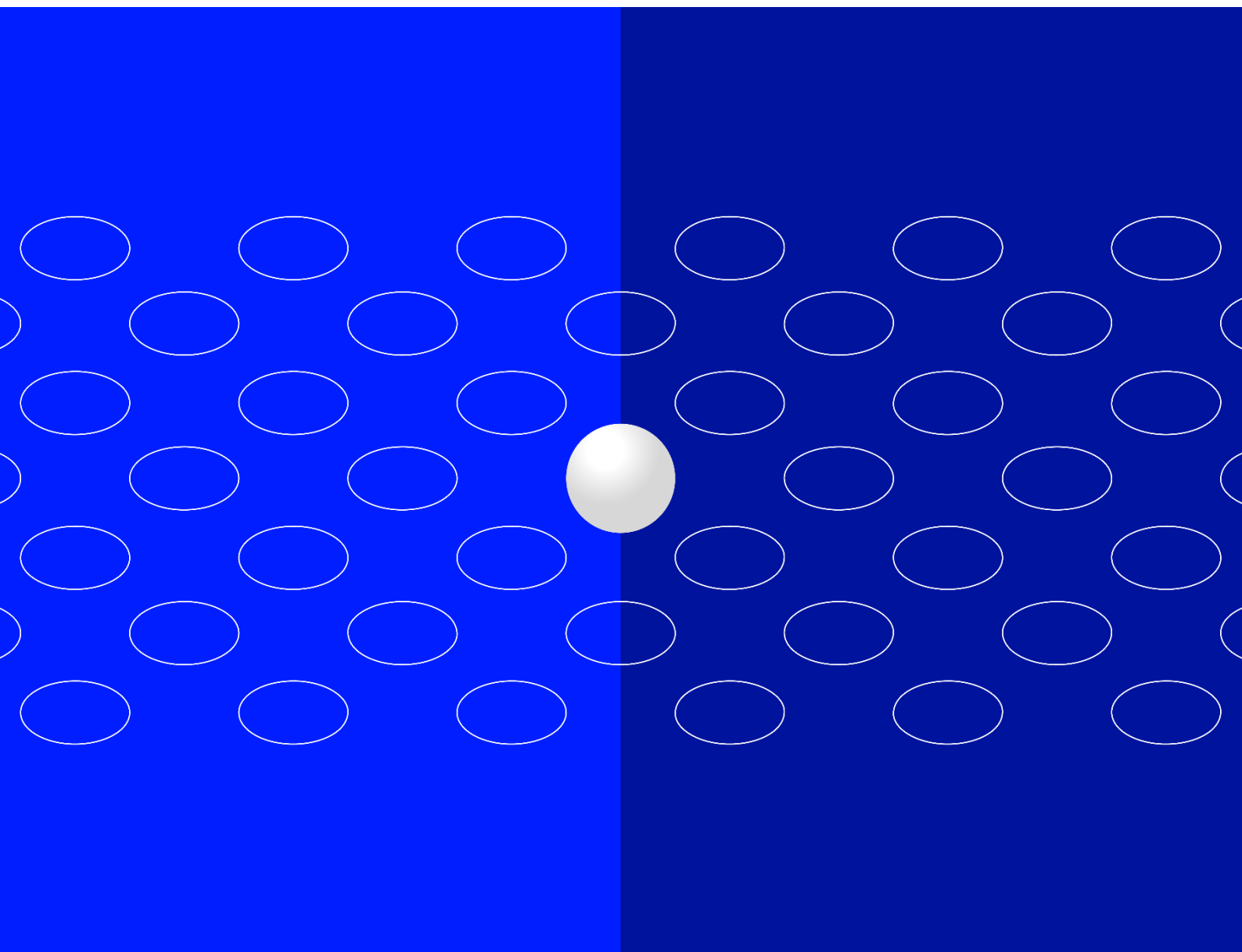


# GROUP TICKER PLANT

LSE GTP 002 – Technical Guide – London Stock Exchange  
ISSUE 26.2

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**LONDON  
STOCK  
EXCHANGE**

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## Guide disclaimer

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# 1 Documentation

## 1.1 This guide

The purpose of this document is to provide a detailed guide to the various message types and message formats as employed by the interface of the Group Ticker Plant. It also provides information on the behaviour and characteristics of our service lines, including Level 1, Level 2 snapshot, Level 2 incremental as supported by the Group Ticker Plant protocol.

This guide will be updated and reissued when appropriate to do so.

## 1.2 Readership

This document is particularly relevant to trading, market data and technical teams within member firms, information vendors and other market participants who currently receive or are interested in receiving London Stock Exchange Group market data.

When read in conjunction with other Group Ticker Plant documents, it is intended to provide all required information to develop to, and interact with, our real-time information systems.

## 1.3 Document series

This guide, 'GTP002 – Technical guide', forms part of the documentation library supporting those clients interacting with the Group Ticker Plant. For information, the full series of currently available documentation is outlined below:

- GTP001 – Product guide
- GTP002 – Technical guide – Turquoise
- **GTP002 – Technical guide – London Stock Exchange (this document)**
- GTP002 – Technical guide – TRADEcho
- GTP003 – Statistics guide
- GTP004 – Parameters guide
- GTP005 – Testing guide

Further documentation to support activities, such as testing and displaying our data, will be released in due course.

The full library of Group Ticker Plant documentation can be found at: <https://www.lseg.com/areas-expertise/technology/group-technology/group-ticker-plant>

## 1.4 Document history

This document has been subject to the following iterations:

| Issue     | Date             | Description                                                                                                                                                                                                      |
|-----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19.9.3.1  | 17 January 2019  | 3.11.24 – Added Trade Summary to Turquoise services.                                                                                                                                                             |
| 19.9.3.2  | 18 January 2019  | Re-branding Turquoise Lit Auctions™ to Turquoise Plato Lit Auctions™. Turquoise Lit Auctions™ has recently become part of the Plato Co-operation agreement.                                                      |
| 19.10     | 26 July 2019     | 3.11.13 – Added Hidden Execution Indicator attribute to Trade message.                                                                                                                                           |
| 19.10.2.5 | 20 January 2020  | 6.2, 6.3.5, 7.2, 3.10.7, 3.11.1, 3.11.2, 3.11.13, 3.11.20 – Added the Turquoise NYLON™ Cash Order Book as a source venue for which Market Data will be published by GTP.                                         |
| 19.10.3   | 9 August 2019    | 3.11.23 – Updated service breakdown with inclusion of EuroTLX Admin channel.<br>6.5 – Added Security Type ML for Milan Bond Market.<br>5 – Updated section due to decommissioning of Tertiary Feed in GTP Milan. |
| 19.10.4   | 23 December 2019 | 3.7 – Off Book Trade message removed. 3.11.3, 3.11.4, 3.11.5, 3.11.6 – Instrument.<br>Directory Messages moved under Application messages.<br>3.11.3, 3.11.5 – Updated Data type of EMS.                         |

| Issue   | Date              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19.11   | 28 February 2020  | <p>3.11.26 – Trade Summary: added Side of resting orders – message becomes applicable to Borsa and EuroTLX venues</p> <p>4.2.2, 4.2.3– Updated Recovery Request of Order book.</p> <p>6.2 – Combined Borsa and EuroTLX in a single Source Venue code.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 19.11.1 | 16 March 2020     | 4.2.2, 4.2.3 – Minor update to Level 1 Order Book Recovery.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 19.11.2 | 14 May 2020       | <p>3.11.14, 3.11.22 – Minor update to footnote and Trade Type definition.</p> <p>3.7 – Updated Intelligent Throttling schema.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 20.2    | 10 August 2020    | <p><b>Changes for Release 1.4.5.0</b></p> <p>3.11.25, 3.11.26, 3.11.27 – Updated MiFID II Trade, MiFID II Trade Report and MiFID II Trade Cross messages positioning of ILQD, SIZE and LRGS flags.</p> <p>3.11.2 – Updated the datatype of the ADT field from Price to Price4.</p> <p>5, 5.1, 5.2, 5.3, 5.4, 5.5 – Sections rewritten to explain new behaviour and details relating to Failure of Primary Feed and Disaster Recovery using Secondary Data Centre.</p> <p>6.9 – Added the tick sizes for the ATF Market.</p> <p>Removed entries on the revision history prior to 2019.</p>                                                                                                                                                                                                                                                                                                                                                                      |
| 20.3    | 30 September 2020 | <p>6.2 – Rebranded Turquoise Global Holdings Limited (TGHL) source venues and added new Turquoise Global Holdings Europe (TGHE) source venues.</p> <p>3.11.2, 3.11.3 – Under the Venue Instrument ID field on the Instrument Directory and Instrument Directory – Equities messages added new Instrument Identifiers that will be used by Instruments of the TGHE source venues.</p> <p>Note: This change will only go live into the production environment in a Hard Brexit Scenario. If and when this does move to production, there will be a specific notice issued to notify customers.</p> <p>3.11.25, 3.11.26 – Added Venue of Publication to distinguish Trade Reports and SI quotes published on EU TRADEcho APA and UK TRADEcho APA.</p> <p>Note: This change will only go live into the production environment in a Hard Brexit Scenario. If and when this does move to production, there will be a specific notice issued to notify customers.</p> |
| 20.4    | 10 January 2021   | <p><b>Changes for Release 1.4.6.0</b></p> <p>All sections – Removed content related to Borsa Italiana and associated venues.</p> <p>3.11.14, 3.11.22 Appended the Trade Qualifier field to the Trade message and the MiFID II Trade message.</p> <p>3.11.22, 3.11.23, 3.11.24 Appended the MMT byte array to the MiFID II Trade, MiFID II Trade Cross and MiFID II Trade Report messages.</p> <p>4.1.3, 4.2.9 Update to how Replay, Recovery and GTP Lite force terminate TCP sessions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 20.5    | 01 February 2021  | L1 Incremental Service is now available for the Turquoise MTF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 20.6    | 26 March 2021     | 4.1.3, 4.2.9 Removed the update to how Replay, Recovery and GTP Lite force terminate TCP sessions with RST flags, instead FIN will be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 20.7    | 01 August 2021    | <p>3.11.27 – Added the Analytics message.</p> <p>3.11.1, 3.11.2, 3.11.4 – Updated the diagram to state that the system event message, instrument status message and the instrument directory messages will be published by the Analytics Gateway.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 20.8    | 15 October 2021   | <p>3.11.15, 3.11.24 – Introduction of BTF trades for Turquoise.</p> <p>6.3.2 – 't' will be used for stating the end of Regular Trading on Turquoise Lit™ and Turquoise Plato Lit Auctions™ Order Books.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 20.9    | 25 May 2022       | 3.11.12 – the 2 <sup>nd</sup> bit will be used to indicate Retail LP orders on Turquoise Plato Lit Auctions™ Order Books                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Issue                           | Date                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
|---------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------|---------------------------------|---------------------------------------------------------|--------------|------------|----------------------|-----------------------------------------------------------------------|-------------------|---------------------------------|
| 20.10                           | 15 September 2022                                                     | 3.11.12 – the note ‘Depth indication won’t be provided for Retail LP orders’ was deleted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 20.11                           | 9 November 2022                                                       | 3.11.23 – Reserved Field (Alpha) is converted into ‘Total Number of Transactions’ (UInt32)<br>4.2.4 – Reject code changed from ‘e’ to ‘a’<br>3.11.17 – Footnote is added to ‘Statistic Type’ field<br>3.11.17 – ‘Not Applicable’ value is added to ‘Auction Info’ field<br>3.11.3 – ‘Price Band Tolerances (%)’ field converted into ‘Reserved Field’                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 20.12                           | 18 January 2023                                                       | Change made with 20.9 was reverted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 20.13                           | 15 February 2023                                                      | Description of rapid login protection is added to section 2.1.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 21.0                            | 20 March 2023                                                         | 3.11.12 – the 2 <sup>nd</sup> bit will be used to indicate Retail LP orders on Turquoise Plato Lit Auctions™ Order Books                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 21.1                            | 5 May 2023                                                            | 3.11.17 – Removed 18,19 statistics types since they are not applicable to LSEG Trading Venues                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 22.0                            | 5 May 2023                                                            | 3.11.17 – Removed the note that Static and Dynamic Reference Prices are not applicable to LSEG Trading Venues. Description of the functionality is added to GTP 003 - Statistics Guide - Issue 11.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 23.0                            | 7 September 2023                                                      | 3.11.14 – Typo in ‘Trade Qualifier’ field is fixed<br>3.11.22 – Typo in ‘Instrument Identification Code Type’ field is fixed<br>3.11.23 - MIFID II Trade Report (0x54) is updated as follow: <ul style="list-style-type: none"><li>• New fields are added: Third-country trading venue of execution, Portfolio Transaction Flag, Contingent Transaction Flag, Missing Price</li><li>• New values are added to the fields: Price Formation Indicator, Reference Price Indicator</li><li>• Description of the following fields is updated: MIFID Price, Price Currency, Notional Currency, Notional Amount</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 23.1                            | 20 October 2023                                                       | 3.11.23 - MIFID II Trade Report (0x54) is updated as follow: <ul style="list-style-type: none"><li>• Description of field ‘MIFID Quantity’ is updated</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| 24                              | 5 January 2024                                                        | Specification has been split across venues (LSE, TradEcho, Turquoise). This document will now contain content applicable to LSE only.<br><br>Section “5. Product failure” was removed from this document and merged with respective section in GTP001 – “4.0 System overview.”<br><br>The following messages are removed: <ul style="list-style-type: none"><li>• Trade Cross</li><li>• Announcements</li><li>• Indicative Quote Information</li><li>• MiFID II Trade Report</li><li>• MiFID II Trade Cross</li><li>• Systematic Internaliser (SI) Quotes</li></ul> <table><thead><tr><th>Message</th><th>Updated description of the fields</th></tr></thead><tbody><tr><td>Instrument Directory – Equities</td><td>Allowed Book Types, Venue Instrument ID, Reserved Field</td></tr><tr><td>System Event</td><td>Event Code</td></tr><tr><td>Instrument Directory</td><td>Allowed Book Types, Venue Instrument ID, Average Daily Turnover (ADT)</td></tr><tr><td>Instrument Status</td><td>Trading Status, Order Book Type</td></tr></tbody></table> | Message | Updated description of the fields | Instrument Directory – Equities | Allowed Book Types, Venue Instrument ID, Reserved Field | System Event | Event Code | Instrument Directory | Allowed Book Types, Venue Instrument ID, Average Daily Turnover (ADT) | Instrument Status | Trading Status, Order Book Type |
| Message                         | Updated description of the fields                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| Instrument Directory – Equities | Allowed Book Types, Venue Instrument ID, Reserved Field               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| System Event                    | Event Code                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| Instrument Directory            | Allowed Book Types, Venue Instrument ID, Average Daily Turnover (ADT) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |
| Instrument Status               | Trading Status, Order Book Type                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |                                   |                                 |                                                         |              |            |                      |                                                                       |                   |                                 |

| Issue                           | Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Add Order MBO Order Book Type                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Add Order MBP Order Book Type                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Add Order Inc Order Book Type, Order Type                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Modify Order Order Book Type                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Delete Order Order Book Type                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | TOB Order Book Type, Flags                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Order Book Clear Order Book Type                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Trade Flags, Hidden Execution Indicator, Trade Qualifier                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Statistics Update Statistic Type, Auction Type, Auction Info, Opening/Closing Price Indicator                                                                                                                                                                                                                                                                                                                                                                        |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Statistics Snapshot Auction Type                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Recovery Request Recovery Type                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MIFID Trade Trade Type, MiFID Price, Instrument Identification Code Type, Price Notation, Price Currency, Notional Currency, PTRefPriceWaiverFlag, Trade Qualifier, Market Mechanism, Trading Mode, Transaction Category, Negotiation Indicator, Agency Cross Indicator, Reference Price Indicator, Special Dividend Indicator, Off Book Automated Indicator, Price Formation Indicator, Post-Trade Deferral Reason, Deferral/Enrichment Type, Duplicative Indicator |
| Message                         | Field is renamed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Instrument Directory – Equities | 'Group ID' into 'Segment'<br>'Flags' into 'Reserved Field'<br>'Security Subtype' into 'Reserved Field'<br>'Settlement System' into 'Reserved Field'<br>'Last Validity Date' into 'Reserved Field'<br>'Settlement Date' into 'Reserved Field'<br>'Venue Underlying ID' into 'Security Exchange'<br>'Underlying ISIN Code' into 'Reserved Field'<br>'Underlying Type' into 'Reserved Field'<br>'Number Of Shares In Circulation' into 'Reserved Field'<br>'Leverage Certificates Barrier' into 'Reserved Field'<br>'Option Style' into 'Reserved Field'<br>'Parity' into 'Reserved Field'<br>'Reserved field' into 'Reserved Field' |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Instrument Directory            | 'Group ID' into 'Segment'<br>'Underlying ISIN Code' into 'Reserved Field'<br>'Underlying Instrument ID' into 'Reserved Field'<br>'Flags' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Add Order MBO                   | 'Yield' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Add Order Short MBO             | 'Yield' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Add Order MBP                   | 'Yield' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Add Order Short MBP             | 'Yield' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TOB                             | 'Bid Yield' into 'Reserved Field'<br>'Offer Yield' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Trade                           | 'Yield' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Statistics Snapshot             | 'Open Interest' into 'Reserved Field'<br>'Volatility' into 'Reserved Field'<br>'TradeHighOffBook' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



| Issue | Date             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                  | 'TradeLowOffBook' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|       |                  | MIFID Trade<br>'Transaction to be Cleared' into 'Reserved Field'<br>'Measurement Unit' into 'Reserved Field'<br>'Quantity in Measurement Unit' into 'Reserved Field'<br>'PTTransTypeFlag' into 'Reserved Field'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|       |                  | <b><u>Release 1.23.3.0 changes:</u></b><br>List of channels before each message is updated in line with the new instrument distribution<br>Trade Summary – new fields Best Bid Size, Best Bid Price, Best Offer Size, Best Offer Price added.<br>3.5 Data Types – Int Size type added, description of Alpha and UDT data types updated.<br>Add Order Incremental – Yield field (offset 44) is reused for Transaction Time field.<br>Order Modify:<br><ul style="list-style-type: none"> <li>• Previous Yield (offset 72) is reused for Transaction Time field,</li> <li>• New yield field (offset 46) is converted to Reserved Field for future use.</li> </ul> Order Delete - Previous Yield (offset 47) is reused for Transaction Time field.<br>MIFID II Trade:<br><ul style="list-style-type: none"> <li>• PTliquidFlag (offset 227) is reused for Market Closing Price Flag (CLSE),</li> <li>• PTDeferralReasonFlag (offset 223) is converted to Reserved Field for future use,</li> <li>• Price Currency (offset 162) – name and description is updated to refer to Major Currency,</li> <li>• Notional Currency (offset 185) – description is updated to refer to Major Currency,</li> <li>• Reference Price Indicator (offset 278) – new value '1' Market Closing Price Trade (CLSE) is added.</li> <li>• Negotiated Indicator (offset 275) – Values N, 1 (NLIQ), 2 (OLIQ), 3 (PRIC) are removed as not applicable for LSE.</li> </ul> |
| 24.1  | 5 January 2024   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 24.2  | 28 February 2024 | 3.11.8 Top of Book – typos in data type column are corrected<br>3.10.5 Replay and Recovery Complete – typo in field length is corrected<br>3.4 Timestamp – added additional clarification related transaction time provided by market various sections - removed information about yield, pending price, implied orders since this functionality is not applicable to LSEG trading venues<br>3.11.9, 3.11.10, 3.11.11 – updated description of field 'Transaction Time'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 24.3  | 24 April 2024    | 3.5 Data Type section – name of extended ASCII table is corrected<br>3.11.4 – Instrument Status - outdated value related to L1 Auction has been removed from field 'Trading Status'<br>3.11.4, 3.11.5, 3.11.7, 3.11.9, 3.11.10, 3.11.12, 3.11.13 - values related to Off-Book and Private RFQ book are removed from field 'Order Book Type' where they are not applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 24.4  | 21 August 2024   | <b>Changes for Release 1.24.0.0</b><br>3.11.2 – Instrument directory<br><ul style="list-style-type: none"> <li>• Reserved field (offset 103) converted to Partition ID</li> <li>• Price Band Tolerances (%) (offset 47) was converted to Reserved Field</li> </ul> 3.11.3 Instrument Directory – Equity<br><ul style="list-style-type: none"> <li>• Reserved field (offset 242) converted to Partition ID</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 25.1  | 11 December 2024 | All sections – Removed content related to FTSE Russell Indices Update<br>3.11.18 - FTSE Russell Indices Update – message removed<br>6.3 – Base36, Base62 decoding alphabets – formulation enhanced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 26.1  | 9 May 2025       | Channel P/p has been added.<br>Sections: 3.9.1, 3.9.2, 3.9.3, 3.10.2, 3.10.3, 3.10.4, 3.10.5, 3.11.1, 3.11.2, 3.11.3, 3.11.4, 3.11.5, 3.11.6, 3.11.7, 3.11.8, 3.11.9, 3.11.10, 3.11.11, 3.11.12, 3.11.13, 3.11.14, 3.11.15, 3.11.16, 3.11.17, 3.11.18, 3.11.19, 3.11.20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Issue | Date            | Description                                                                                                                                                                                                                                                 |
|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                 | <b>Changes for Release 1.25.3.0</b>                                                                                                                                                                                                                         |
|       |                 | 3.11.2 'Instrument Directory – Equities' (0x52) / 3.11.3 'Instrument Directory' (0x70) – a new value — 4 — will now appear in the field Partition ID in addition to values 1, 2, and 3. This will apply to instruments that are moved to the 4th partition. |
|       |                 | 5.1 Security Types – the section has been updated and outdated values have been removed.                                                                                                                                                                    |
| 26.2  | 15 October 2025 | <b><u>FCA PS 24/14 implementation:</u></b>                                                                                                                                                                                                                  |
|       |                 | <b>3.11.18 MiFID II Trade (0x51) –</b> MiFID Quantity will be blank if the traded security is a bond.                                                                                                                                                       |
|       |                 | To determine the asset class of an instrument, clients should refer to:                                                                                                                                                                                     |
|       |                 | <ul style="list-style-type: none"> <li>• MIT401 – Guide to Reference Data Services, OR</li> <li>• The code published in offset 165 of 'Instrument Directory – Equity' (0x52). The meaning of the code is provided in section 5.1 Security Types.</li> </ul> |

In subsequent issues of this document, where amendments have been made, these changes will be indicated through the use of **red text**.

## 1.5 Enquiries

For further information on Group Ticker Plant, please contact either your Technical Account Manager or the Client Technology Group (UK):

- Telephone: +44 (20)7797 3939
- E-mail: [londontam@lseg.com](mailto:londontam@lseg.com)

Please contact the market data team of LSEG for further information regarding data licensing for this service by e-mailing: [marketdata@lseg.com](mailto:marketdata@lseg.com).

Further information can also be found on our project websites:

<https://www.lseg.com/areas-expertise/technology/group-technology/group-ticker-plant>

## 2 Connectivity

### 2.1 Transmission Standards

The Group Ticker Plant employs industry standard data delivery and recovery transmission techniques. Further details are provided below.

Clients should note that network addressing is included as part of GTP004 – Parameters guide.

#### 2.1.1 Multicast Channels

The Group Ticker Plant delivers real-time market data over a number of load-balanced IP multicast channels. Details of the service line allocation across multicast channels can be found in GTP001 – Product guide.

The real-time channels transmit in UDP network packets over IP version 4 (IPv4) Ethernet standards. UDP header information is as defined in the IETF RFC 791 (IPv4) and RFC 768 (UDP) transmission standards. While each UDP network packet will contain a single Unit Header, multiple application messages may be packaged in a single network packet. This is done in an effort to manage client bandwidth requirements. Further details are contained in section 4.3.1 of GTP001 – Product guide.

Clients should subscribe to both primary and secondary market data feeds. While, during normal service, replay and recovery services are only available on the primary market data gateway, through subscription to both primary and secondary feeds clients are able to arbitrage messages – recovering any missed messages on the primary market data feed from the secondary market data feed.

#### 2.1.2 Recovery and Replay Services

The Recovery and Replay unicast channels will guarantee data delivery through use of TCP over IP version 4 (IPv4) Ethernet standards. TCP Header information is defined in the IETF RFC 793 standard and IPv4 is defined in the RFC 791 standard.

A protection mechanism is in place to protect Recovery and Replay gateways from rapid logins/logouts/disconnects. A rapid login/logout/disconnect is one that occurs very quickly following a previous login/logout/disconnect. Once a user has triggered this mechanism, the gateway response to each successive login/logout attempt or disconnect will be subject to a delayed response that grows exponentially. It is recommended that all users maintain a 100mS interval between successive login or logout attempts and disconnects to each gateway (including where the gateway has initiated the disconnection) to keep a safe margin and avoid accidentally triggering this protection mechanism.

### 2.2 Client Identification

All clients are required to request enablement on Group Ticker Plant service lines ahead of access to either CDS or Production environments. Access is permissioned on both the Group Ticker Plant gateways and at our network firewalls.

Upon successful completion of access request, the Group will allocate clients their CompID(s). The CompID should be used by clients to log in to either the replay or recovery services. Each CompID will be permissioned to access all Group Ticker Plant multicast gateways for which a client has the correct commercial agreement(s) in place, each CompID can only be logged in to one replay or recovery service at any one time.

Unlike the existing Millennium Exchange market data solution, the Group Ticker Plant does not require a password to access the replay or recovery services. The removal of password validation was in direct response to client feedback on our existing product provision.

## 3 Message formats

Group Ticker Plant delivers all real-time multi-asset class information in a single, bespoke, binary protocol. In close collaboration with clients, the Group worked to build an improved library of messages which are intuitive and clear; we believe we have made significant improvements to our legacy market data protocols. Some of these improvements include:

- Removal of password validation on both replay and recovery services
- Removal of logout process from both replay and recovery services
- Decommission of the Level 2-MITCH Time message, replacing with nanosecond time-stamping on all application messages
- Collapse of Level 2-MITCH Auction Trade, Trade, Trade Break and Order Executed messages into a single Trade message
- Introduction of the Statistics and the Statistics Update message to support the publication of derived information
- Introduction of asset-specific Instrument Directory messages available on the recovery solution to support the publication of instrument-specific reference data
- Inclusion of the 'Allowed Book Types' field in the Instrument Directory messages to facilitate the identification of applicable trading models for an instrument
- Introduction of the Top of Book message to support our new real-time streaming level 1 service. This includes a flag to indicate additional executable depth below the Best Bid and Offer

The Group protocol has been developed to be as generic as possible, facilitating the introduction of additional services as we develop and enhance the Group Ticker Plant product. In some cases, due to differences in trading models of the source systems, it was not possible to maintain a generic implementation, suitable for all systems.

### 3.1 Packet composition

The Unit Header is used to deliver all administrative and application messages to and from the server on all multicast and unicast channels. Whilst a Unit Header may contain multiple application messages, it will never contain more than one administrative message. A Unit Header will not contain a combination of both application and administrative messages.

### 3.2 Message types

The Group Ticker Plant broadcasts a library of messages. The messages are categorised as either administrative or application messages. The administrative messages are used on the TCP/IP replay and recovery and GTP Lite services to implement the interaction of a user with the system. The application messages are used to broadcast our real-time data service lines on the multicast channels, and to provide certain static reference data information via replay and recovery services.

Clients should treat each application message as a single standalone instruction, updating their order books and systems appropriately based on the content of the application message. Clients should not program to a multicast stream of messages in an attempt to identify patterns or system logics, as the stream of messages disseminated is subject to change as we optimise our trading and market data technologies. If a client processes each application message in real-time as a standalone instruction, order books will be a true state of the trading engines of the supported markets.

### 3.3 Sequence Numbers

All application messages transmitted by the server on the multicast channels and the replay services are sequenced. The Unit Header only contains the sequence number of the first message; the sequence numbers of any other messages included in the same packet are implied. The sequence number of the next packet can be determined by adding the value in the Message Count field of the Unit Header to the value in the Sequence Number field.

The application messages sent from the server by the recovery service and all administrative messages (including those sent by the client) are un-sequenced.

### 3.4 Timestamps

All Group Ticker Plant application messages contain a timestamp with nanosecond granularity and will be sent in UTC. Timestamps are derived from the supporting infrastructure timestamp, which is synched through various processes including industry standards such as NTP – guaranteeing accuracy to the millisecond – and PTP allowing accuracy to microsecond.

Transaction times included in Trade, Add Order Incremental, Order Modify, Order Delete messages will be disseminated as reported to the Group Ticker Plant by upstream systems. Where granularity to nanosecond does not exist, the timestamp will

be rounded to the nearest microsecond and disseminated to clients. It is assumed that each upstream system, by adhering to the level of accuracy requested in MIFID II RTS25 Table 2, will ensure the appropriate timestamp granularity is transmitted to Group Ticker Plant.

The upstream system's transaction time is consistent across all interfaces – GTP market data, trading, drop copy and post trade interfaces.

For more details on transaction time, clients should refer to section "Level 2 incremental service" in "GTP 001 – Group Ticker Plant Product Guide".

### 3.5 Data types

The fields of the messages utilized by the server will support the data types outlined below:

| Data type            | Length   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alpha                | Variable | These fields use standard ASCII and Windows-1252 (CP-1252) based on Latin-1 (ISO8859-1) character bytes. They are left justified and padded on the right with spaces.                                                                                                                                                                                                                                                                                                                                                                                 |
| Bit Field            | 1        | A single byte used to hold up to eight 1-bit flags. Each bit will represent a Boolean flag. The 0 bit is the lowest significant bit and the 7 bit is the highest significant bit.                                                                                                                                                                                                                                                                                                                                                                     |
| Byte                 | 1        | A single byte used to hold one ASCII character.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Date                 | 8        | Date specified in the YYYYMMDD format using ASCII characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Time                 | 6        | Time specified in HHMMSS format using ASCII characters in a 24-hour clock format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| UDT (Unix Date Time) | 8        | Little-Endian encoded 64 bit unsigned integer where; time stamp (in UTC) = (date time per second resolution in unix time format) * 1,000,000,000 + (nanoseconds component).                                                                                                                                                                                                                                                                                                                                                                           |
| Price                | 8        | Signed Little-Endian encoded 64 bit integer field with eight implied decimal places.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Size                 | 8        | Little-Endian encoded 64 bit unsigned integer with eight implied decimal places.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Price4               | 8        | Signed Little-Endian encoded 64 bit integer field with four implied decimal places.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Size4                | 8        | Little-Endian encoded 64 bit unsigned integer with four implied decimal places.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| UInt8                | 1        | 8 bit unsigned integer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| UInt16               | 2        | Little-Endian encoded 16 bit unsigned integer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| UInt32               | 4        | Little-Endian encoded 32 bit unsigned integer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| UInt64               | 8        | Little-Endian encoded 64 bit unsigned integer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Date and Time        | 27       | ISO 8601 date and time in the following string format:YYYY-MM-DDThh:mm:ss.dddZ.<br>– 'YYYY' is the year<br>– 'MM' is the month<br>– 'DD' is the day<br>– 'T' – means that the letter 'T' shall be used<br>– 'hh' is the hour<br>– 'mm' is the minute<br>– 'ss.ddd' is the second and its fraction of a second<br>– Z is UTC time<br>Dates and times shall be reported in UTC.                                                                                                                                                                         |
| MiFID Decimal        | 20       | These fields use standard ASCII character bytes to represent numeric values. They are left justified and padded on the right with spaces.<br>{DECIMAL-n/m} – Decimal number of up to 'n' digits in total of which up to 'm' digits can be fraction digits. Decimal separator is '.' (Full stop). Negative numbers are prefixed with '-' (minus).<br>Where applicable, values shall be rounded and not truncated.<br>The default value that is populated when there is no valid value can be '0' (zero) or 'spaces', dependent on the upstream system. |

| Data type | Length | Description                                                                          |
|-----------|--------|--------------------------------------------------------------------------------------|
| Int Size  | 8      | Signed Little-Endian encoded 64 bit integer field with eight implied decimal places. |

Please note that some field descriptions in this document include 'blank' as valid values. 'Blank' should be considered as 'space filled' for Alpha data types and '0' (zero) for data types Byte, Price, Size, UInt8, UInt16, UInt32 and UInt64.

### 3.6 Message overview – Administrative Messages

| Name                         | Message type |      | Usage                                                                                                          |
|------------------------------|--------------|------|----------------------------------------------------------------------------------------------------------------|
|                              | ASCII        | Hex  |                                                                                                                |
| Heartbeat                    | -            | -    | Used by the server, on the real-time service, to exercise the communication line during periods of inactivity. |
| Login Request                | (soh)        | 0x01 | Used by the client to log in to the replay or recovery channel.                                                |
| Login Response               | (stx)        | 0x02 | Used by the server to accept or reject a login request to the replay or recovery channel.                      |
| Replay Request               | (etx)        | 0x03 | Used by the client to request a retransmission of messages on the replay channel.                              |
| Replay Response              | (eot)        | 0x04 | Used by the server to respond to a retransmission request on the replay channel.                               |
| Recovery Request             | •            | 0x81 | Used by the client to request data on the recovery channel.                                                    |
| Recovery Response            | ,            | 0x82 | Used by the server to respond to a snapshot request on the Snapshot channel.                                   |
| Replay and Recovery Complete | <i>f</i>     | 0x83 | Used by the server to indicate the successful completion of servicing a message replay or a recovery request.  |

### 3.7 Message overview – Application Messages

Each GTP Application Message is listed below with corresponding ASCII and Hex codes. Intelligent Throttling configuration is indicated for Level 2 Incremental (L2I) and Level 2 Snapshot (L2S) services. For additional information related to Intelligent Throttling, please refer to GTP Product Guide (GTP001).

| Name                            | Message type |      | Usage                                                                                                                                                           | Intelligent Throttling applied? |     |
|---------------------------------|--------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----|
|                                 | ASCII        | Hex  |                                                                                                                                                                 | London                          |     |
|                                 |              |      |                                                                                                                                                                 | L2I                             | L2S |
| System Event                    | S            | 0x53 | Sent to indicate the start and end of the day.                                                                                                                  | N                               | N   |
| Instrument Directory            | p            | 0x70 | Used to disseminate a common and limited set of data for all configured instrument types (except strategy instruments) on the real-time channels.               | Y                               | Y   |
| Instrument Directory – Equities | R            | 0x52 | Used to disseminate reference data information of equity instruments.                                                                                           |                                 |     |
| Instrument Status               | H            | 0x48 | Used to communicate scheduled and unscheduled session changes. When sent in the recovery channel, used to indicate the current trading status of an instrument. | N                               | N   |
| Add Order                       | A            | 0x41 | Indicates the first order of a given side of an MBO snapshot.                                                                                                   |                                 | Y   |
| Add Order Short                 | e            | 0x65 | Used to indicate individual orders of an MBO snapshot.                                                                                                          |                                 | Y   |
| Add Order MBP                   | f            | 0x66 | Indicates the first price point of a given side of an MBP snapshot.                                                                                             |                                 | Y   |
| Add Order Short MBP             | g            | 0x67 | Used to indicate individual price points of an MBP snapshot.                                                                                                    |                                 | Y   |
| Add Order Incremental           | F            | 0x46 | Sent to instruct recipients to add a new displayable order to the retrospective order book.                                                                     | N                               |     |
| Delete Order                    | D            | 0x44 | Sent to instruct recipients to delete an order from the retrospective order book.                                                                               | N                               |     |
| Modify Order                    | U            | 0x55 | Sent to instruct recipients to update an order's price and/or size on the retrospective order book.                                                             | N                               |     |
| Top of Book                     | i            | 0x69 | Used to update the level 1 service following any change to the consolidated Best Bid and Offer.                                                                 |                                 | N   |
| Order Book Clear                | y            | 0x79 | Sent to instruct recipients to remove all orders from the order book for the specified instrument.                                                              | N                               | N   |
| Trade                           | P            | 0x50 | Sent to indicate trades executed on supported markets.                                                                                                          | N                               | N   |
| Statistics                      | w            | 0x77 | Contains a set of statistics that are updated frequently, usually as a result of executions.                                                                    | N                               | N   |
| Statistics Update               | j            | 0x6A | Contains a set of statistics that are not updated frequently.                                                                                                   | N                               | N   |
| Statistics Snapshot             | k            | 0x6b | A snapshot of an instrument's statistics that is used for recovery.                                                                                             |                                 |     |

| Name          | Message type |      | Usage                                                                                                    | Intelligent Throttling applied? |     |
|---------------|--------------|------|----------------------------------------------------------------------------------------------------------|---------------------------------|-----|
|               | ASCII        | Hex  |                                                                                                          | London                          |     |
|               |              |      |                                                                                                          | L2I                             | L2S |
| MiFID Trade   | Q            | 0x51 | Sent to represent different types of MiFID compliant trades published by markets.                        | N                               | N   |
| Trade Summary | W            | 0x57 | Publishing Trade Summaries for both Multi- and Single-Fill Trades                                        | N                               |     |
| Analytics     | a            | 0x61 | Analytics Message is used to disseminate additional statistics including order book activity statistics. |                                 |     |

### 3.8 Unit header

| Field             | Offset | Length   | Type   | Description                                                                |
|-------------------|--------|----------|--------|----------------------------------------------------------------------------|
| Length            | 0      | 2        | UInt16 | Length of the message block including the header and all payload messages. |
| Message Count     | 2      | 1        | UInt8  | Number of payload messages that will follow the header.                    |
| Market Data Group | 3      | 1        | Byte   | Identity of the market data group the payload messages relate to.          |
| Sequence Number   | 4      | 4        | UInt32 | Sequence number of the first payload message.                              |
| Payload           | 8      | Variable | -      | One or more payload messages.                                              |



## 3.9 Administrative Messages (Client-Initiated)

### 3.9.1 Login Request

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field        | Offset | Length | Type   | Description                             |
|--------------|--------|--------|--------|-----------------------------------------|
| Length       | 0      | 2      | UInt16 | Length of message including this field. |
| Message Type | 2      | 1      | Byte   | <b>Hex</b> <b>Meaning</b>               |
|              |        |        |        | 0x01      Login Request                 |
| Username     | 3      | 8      | Alpha  | CompID assigned to the client.          |

### 3.9.2 Replay Request

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | N        |

| Field         | Offset | Length | Type   | Description                                                                                                                               |
|---------------|--------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Length        | 0      | 2      | UInt16 | Length of message including this field.                                                                                                   |
| Message Type  | 2      | 1      | Byte   | <div>Hex</div> <div>Meaning</div>                                                                                                         |
|               |        |        |        | 0x03      Replay Request                                                                                                                  |
| First Message | 0      | 2      | UInt32 | Sequence number of the first message in range to be retransmitted.                                                                        |
| Count         | 7      | 4      | UInt32 | Number of messages to be resent.                                                                                                          |
| Request ID    | 11     | 4      | UInt32 | The value set in this will be echoed back in the corresponding Replay Response. The system will not validate uniqueness of the set value. |

### 3.9.3 Recovery Request

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | Y        |

| Field         | Offset | Length | Type   | Description                                                    |                   |
|---------------|--------|--------|--------|----------------------------------------------------------------|-------------------|
| Length        | 0      | 2      | UInt16 | Length of message including this field.                        |                   |
| Message Type  | 2      | 1      | Byte   | <b>Hex</b>                                                     | <b>Meaning</b>    |
|               |        |        |        | 0x81                                                           | Recovery Request  |
| Request Level | 3      | 1      | UInt8  | Defines the level of the request:                              |                   |
|               |        |        |        | <b>Hex</b>                                                     | <b>Meaning</b>    |
|               |        |        |        | 0                                                              | Instrument        |
|               |        |        |        | 1                                                              | Group (Segment)   |
|               |        |        |        | 2                                                              | Multicast Channel |
| Instrument    | 4      | 8      | UInt64 | GTP Instrument identifier if Request Level is 0. Blank if not. |                   |
| Group ID      | 12     | 6      | Alpha  | Group/Segment ID if Request Level is 1. Blank if not.          |                   |

| Field           | Offset       | Length | Type   | Description                                                                                                                                                                                                                                                                          |                       |
|-----------------|--------------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Order Book Type | 18           | 1      | UInt8  | Only considered if the Request Level is 0. If specified, only data related to the specified order book type is provided. If not specified, data for all available book types for the instrument are provided.<br>For Recovery Type = 3 (Statistics) this has to be set to '0' = All. |                       |
|                 |              |        |        | Value                                                                                                                                                                                                                                                                                | Meaning               |
|                 |              |        |        | 0                                                                                                                                                                                                                                                                                    | All Books             |
|                 |              |        |        | 1                                                                                                                                                                                                                                                                                    | Firm Quote Book       |
|                 |              |        |        | 2                                                                                                                                                                                                                                                                                    | Off-book              |
|                 |              |        |        | 3                                                                                                                                                                                                                                                                                    | Electronic Order Book |
|                 |              |        |        | 4                                                                                                                                                                                                                                                                                    | Private RFQ           |
| Source Venue    | 19           | 2      | UInt16 | Mandatory field if Request Level = 1. Not considered for other Request Levels.                                                                                                                                                                                                       |                       |
|                 |              |        |        | Value                                                                                                                                                                                                                                                                                | Meaning               |
|                 |              |        |        | 1                                                                                                                                                                                                                                                                                    | London Stock Exchange |
| Recovery Type   | 21           | 1      | UInt8  | The type of messages to be replayed:                                                                                                                                                                                                                                                 |                       |
|                 |              |        |        | Value                                                                                                                                                                                                                                                                                | Meaning               |
|                 |              |        |        | 0                                                                                                                                                                                                                                                                                    | Instrument Directory  |
|                 |              |        |        | 1                                                                                                                                                                                                                                                                                    | Order book            |
|                 |              |        |        | 2                                                                                                                                                                                                                                                                                    | All Trades            |
|                 |              |        |        | 3                                                                                                                                                                                                                                                                                    | Statistics            |
|                 |              |        |        | 4                                                                                                                                                                                                                                                                                    | Instrument Status     |
|                 |              |        |        | 5                                                                                                                                                                                                                                                                                    | Reserved              |
| 6               | System Event |        |        |                                                                                                                                                                                                                                                                                      |                       |
| Sequence Number | 22           | 4      | UInt32 | Only valid if Recovery Type = 2 (Trades). If specified, the trades reported with an equal or higher sequence number will be sent.                                                                                                                                                    |                       |
| Request ID      | 26           | 4      | UInt32 | The value set in this will be echoed back in the corresponding Recovery Response and Recovery Complete. The system will not validate uniqueness of the set value.                                                                                                                    |                       |

## 3.10 Administrative Messages (Server-Initiated)

### 3.10.1 Heartbeat

A Unit Header with a Message Count of zero (0) will be used by the server as the Heartbeat message. Such a message will never increment the sequence number of the real-time multicast channel. The next anticipated sequence number will be included in the Sequence Number to enable recipients to detect gaps on the real-time channel.

### 3.10.2 Login response

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field        | Offset | Length | Type   | Description                              |                                   |
|--------------|--------|--------|--------|------------------------------------------|-----------------------------------|
| Length       | 0      | 2      | UInt16 | Length of message, including this field. |                                   |
| Message Type | 2      | 1      | Byte   | Hex                                      | Meaning                           |
|              |        |        |        | 0x02                                     | Login Response                    |
| Status       | 3      | 1      | Byte   | Status of the login request.             |                                   |
|              |        |        |        | Value                                    | Meaning                           |
|              |        |        |        | A                                        | Login Accepted                    |
|              |        |        |        | a                                        | CompID Inactive/Suspended         |
|              |        |        |        | b                                        | Login Limit Reached               |
|              |        |        |        | c                                        | Service Unavailable               |
|              |        |        |        | d                                        | Maximum Connections Limit Reached |
|              |        |        |        | e                                        | Failed (other)                    |
|              |        |        |        | f                                        | Invalid CompID or IP Address      |

### 3.10.3 Replay Response

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | N        |

| Field         | Offset | Length | Type   | Description                                                                                                                       |                          |
|---------------|--------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Length        | 0      | 2      | UInt16 | Length of message including this field.                                                                                           |                          |
| Message Type  | 2      | 1      | Byte   | Hex                                                                                                                               | Meaning                  |
|               |        |        |        | 0x04                                                                                                                              | Replay Response          |
| First Message | 3      | 4      | UInt32 | Sequence number of the first message in range to be retransmitted. This will be zero if Status is not "A".                        |                          |
| Count         | 7      | 4      | UInt32 | Number of messages to be resent, not including any Replay and Recovery Complete messages. This will be zero if Status is not "A". |                          |
| Status        | 11     | 1      | Byte   | Value                                                                                                                             | Meaning                  |
|               |        |        |        | A                                                                                                                                 | Request Accepted         |
|               |        |        |        | D                                                                                                                                 | Request Limit Reached    |
|               |        |        |        | O                                                                                                                                 | Out of Range             |
|               |        |        |        | U                                                                                                                                 | Replay Unavailable       |
|               |        |        |        | c                                                                                                                                 | Concurrent Limit Reached |
|               |        |        |        | e                                                                                                                                 | Failed (Other)           |
| Request ID    | 12     | 4      | UInt32 | Will include the value set as Request ID in the Replay Request message.                                                           |                          |

## 3.10.4 Recovery Response

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | Y        |

| Field           | Offset | Length | Type   | Description                                                                                                                                                                           |                                        |
|-----------------|--------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Length          | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                                               |                                        |
| Message Type    | 2      | 1      | Byte   | Hex                                                                                                                                                                                   | Meaning                                |
|                 |        |        |        | 0x82                                                                                                                                                                                  | Replay Response                        |
| Sequence Number | 3      | 4      | UInt32 | Sequence number on the real-time channel with which the snapshot is synchronised when Recovery Type = 1, 2, 3, 4. This will be zero for other Recovery Types or if Status is not “A”. |                                        |
| Count           | 7      | 4      | UInt32 | Number of messages to follow, not including any Replay and Recovery Complete messages. This will be zero if Status is not “A”.                                                        |                                        |
| Status          | 11     | 1      | Byte   | Value                                                                                                                                                                                 | Meaning                                |
|                 |        |        |        | A                                                                                                                                                                                     | Request Accepted                       |
|                 |        |        |        | O                                                                                                                                                                                     | Out of Range                           |
|                 |        |        |        | a                                                                                                                                                                                     | Invalid Group or Instrument            |
|                 |        |        |        | b                                                                                                                                                                                     | Request Limit Reached                  |
|                 |        |        |        | c                                                                                                                                                                                     | Concurrent Limit Reached               |
|                 |        |        |        | d                                                                                                                                                                                     | Invalid Recovery Type or Request Level |
|                 |        |        |        | e                                                                                                                                                                                     | Failed (Other)                         |
| Request ID      | 12     | 4      | UInt32 | Will include the value set as Request ID in the Recovery Request message.                                                                                                             |                                        |

### 3.10.5 Replay and Recovery Complete

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field          | Offset | Length | Type   | Description                                                                                                                                               |
|----------------|--------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length         | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                   |
| Message Type   | 2      | 1      | Byte   | <div>Hex</div> <div>Meaning</div>                                                                                                                         |
|                |        |        |        | <div>0x83</div> <div>Replay and Recovery Complete</div>                                                                                                   |
| Request ID     | 3      | 4      | UInt32 | Will include the value set as Request ID in the Recovery Request message.                                                                                 |
| Trading Status | 7      | 1      | Byte   | Current Trading status of the Instrument. Populated only when the message is sent at the end of individual order book snapshots during a trading session. |



## 3.11 Application Messages

### 3.11.1 System Event

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field        | Offset | Length | Type   | Description                                                  |                       |
|--------------|--------|--------|--------|--------------------------------------------------------------|-----------------------|
| Length       | 0      | 2      | UInt16 | Length of message including this field.                      |                       |
| Message Type | 2      | 1      | Byte   | Hex                                                          | Meaning               |
|              |        |        |        | 0x53                                                         | System Event          |
| Timestamp    | 3      | 8      | UDT    | Time the message was generated.                              |                       |
| Event Code   | 11     | 1      | Byte   | Value                                                        | Meaning               |
|              |        |        |        | C                                                            | End of Day            |
|              |        |        |        | O                                                            | Start of Day          |
| Source Venue | 12     | 2      | UInt16 | Venue from which market data is received for the instrument. |                       |
|              |        |        |        | Value                                                        | Meaning               |
|              |        |        |        | 1                                                            | London Stock Exchange |

## 3.11.2 Instrument directory

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | N        |

| Field              | Offset | Length | Type      | Description                                                                                                                                                |                       |
|--------------------|--------|--------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Length             | 0      | 2      | UInt16    | Length of message including this field.                                                                                                                    |                       |
| Message Type       | 2      | 1      | Byte      | Hex                                                                                                                                                        | Meaning               |
|                    |        |        |           | 0x70                                                                                                                                                       | Instrument Directory  |
| Timestamp          | 3      | 8      | UDT       | Time the message was generated.                                                                                                                            |                       |
| Instrument         | 11     | 8      | UInt64    | GTP Instrument identifier.                                                                                                                                 |                       |
| ISIN               | 19     | 12     | Alpha     | ISIN Code of an instrument.                                                                                                                                |                       |
| Allowed Book Types | 31     | 1      | Bit Field | Defines the order-book types that are allowed for the instrument. Each designated bit represents a book type. '0' means not allowed and '1' means allowed: |                       |
|                    |        |        |           | Bit                                                                                                                                                        | Name                  |
|                    |        |        |           | 1                                                                                                                                                          | Firm Quote Book       |
|                    |        |        |           | 2                                                                                                                                                          | Off-book              |
|                    |        |        |           | 3                                                                                                                                                          | Electronic Order Book |
|                    |        |        |           | 4                                                                                                                                                          | Private RFQ           |
| Source Venue       | 32     | 2      | UInt16    | Venue from which market data is received for the instrument.                                                                                               |                       |
|                    |        |        |           | Value                                                                                                                                                      | Meaning               |
|                    |        |        |           | 1                                                                                                                                                          | London Stock Exchange |

| Field                                  | Offset | Length | Type      | Description                                                                                                                         |
|----------------------------------------|--------|--------|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| Venue Instrument ID                    | 34     | 11     | Alpha     | Instrument identifier used by the source venue.                                                                                     |
| Tick ID                                | 45     | 2      | Alpha     | The tick structure applicable for the instrument.                                                                                   |
| Reserved Field                         | 47     | 8      | Price     | Reserved for future use.                                                                                                            |
| Dynamic Circuit Breaker Tolerances (%) | 55     | 8      | Price     | Dynamic Circuit Breaker Tolerance (%) of the instrument.                                                                            |
| Static Circuit Breaker Tolerances (%)  | 63     | 8      | Price     | Static Circuit Breaker Tolerance (%) of the instrument.                                                                             |
| Segment                                | 71     | 6      | Alpha     | Segment the instrument is assigned to.                                                                                              |
| Reserved Field                         | 77     | 12     | Alpha     | Reserved for future use                                                                                                             |
| Reserved Field                         | 89     | 11     | Alpha     | Reserved for future use                                                                                                             |
| Currency                               | 100    | 3      | Alpha     | Currency Code as per ISO 4217.<br>For additional currencies supported refer to the Additional Field Values section of this document |
| Partition ID                           | 103    | 1      | Byte      | Trading System's partition in which the instrument is traded                                                                        |
| Reserved Field                         | 104    | 4      | Alpha     | Reserved for future use.                                                                                                            |
| Average Daily Turnover (ADT)           | 108    | 8      | Price4    | Not Applicable to LSE                                                                                                               |
| Reserved Field                         | 116    | 8      | Alpha     | Reserved for future use.                                                                                                            |
| Reserved Field                         | 124    | 1      | Bit Field | Reserved for future use                                                                                                             |
| Reserved Field                         | 125    | 8      | Price     | Reserved for future use.                                                                                                            |
| Reserved Field                         | 133    | 8      | Price     | Reserved for future use.                                                                                                            |

## 3.11.3 Instrument Directory – Equities

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | Y        |

| Field              | Offset | Length | Type      | Description                                                                                                                                            |                                 |
|--------------------|--------|--------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Length             | 0      | 2      | UInt16    | Length of message including this field.                                                                                                                |                                 |
| Message Type       | 2      | 1      | Byte      | Hex                                                                                                                                                    | Meaning                         |
|                    |        |        |           | 0x52                                                                                                                                                   | Instrument Directory – Equities |
| Timestamp          | 3      | 8      | UDT       | Time the message was generated.                                                                                                                        |                                 |
| Instrument         | 11     | 8      | UInt64    | GTP Instrument identifier.                                                                                                                             |                                 |
| ISIN               | 19     | 12     | Alpha     | ISIN code of the instrument.                                                                                                                           |                                 |
| SEDOL              | 31     | 8      | Alpha     | SEDOL code of the instrument.                                                                                                                          |                                 |
| Allowed Book Types | 39     | 1      | Bit Field | Defines the order-book types that are allowed for the instrument. Each designated bit represents a book type. 0 Means not allowed and 1 means allowed: |                                 |
|                    |        |        |           | Bit                                                                                                                                                    | Name                            |
|                    |        |        |           | 1                                                                                                                                                      | Firm Quote Book                 |
|                    |        |        |           | 2                                                                                                                                                      | Off-book                        |
|                    |        |        |           | 3                                                                                                                                                      | Electronic Order Book           |
|                    |        |        |           | 4                                                                                                                                                      | Private RFQ                     |

| Field                                  | Offset | Length | Type      | Description                                                                                                                                                      |                       |
|----------------------------------------|--------|--------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Source Venue                           | 40     | 2      | UInt16    | Venue from which market data is received for the instrument.                                                                                                     |                       |
|                                        |        |        |           | Value                                                                                                                                                            | Meaning               |
|                                        |        |        |           | 1                                                                                                                                                                | London Stock Exchange |
| Venue Instrument ID                    | 42     | 11     | Alpha     | Instrument identifier used by the source venue.                                                                                                                  |                       |
| Segment                                | 53     | 6      | Alpha     | Segment the instrument is assigned to.                                                                                                                           |                       |
| Currency                               | 59     | 3      | Alpha     | Currency code as per ISO 4217.<br><br>For additional currencies supported refer to the Additional Field Values section of this document.                         |                       |
| Tick ID                                | 62     | 2      | Alpha     | The tick structure applicable for the instrument.                                                                                                                |                       |
| Previous Day's Closing Price           | 64     | 8      | Price     | Closing price reported for the previous trading day.                                                                                                             |                       |
| Reserved Field                         | 72     | 8      | Price     | Reserved for future use                                                                                                                                          |                       |
| Dynamic Circuit Breaker Tolerances (%) | 80     | 8      | Price     | Dynamic Circuit Breaker Tolerance (%) of the instrument.                                                                                                         |                       |
| Static Circuit Breaker Tolerances (%)  | 88     | 8      | Price     | Static Circuit Breaker Tolerance (%) of the instrument.                                                                                                          |                       |
| Reserved Field                         | 96     | 1      | Bit Field | Reserved for future use                                                                                                                                          |                       |
| Reserved Field                         | 97     | 1      | UInt8     | Reserved for future use                                                                                                                                          |                       |
| Expiration Date                        | 98     | 8      | Date      | Expiration date of the instrument.                                                                                                                               |                       |
| Listing Start Date                     | 106    | 8      | Date      | Listing start date of the instrument.                                                                                                                            |                       |
| Listing End Date                       | 114    | 8      | Date      | Listing start date of the instrument.                                                                                                                            |                       |
| Minimum Lot/Minimum Execution Size     | 122    | 8      | Size      | Indicates the minimum quantity/nominal value tradable on the market for a security.                                                                              |                       |
| Last Price In Preceding Session        | 130    | 8      | Price     | Last execution price in a session prior to the current trading day.                                                                                              |                       |
| Last Price In Preceding Session Date   | 138    | 8      | Date      | Last execution date in a session prior to current trading day.                                                                                                   |                       |
| Reserved Field                         | 146    | 1      | UInt8     | Reserved for future use                                                                                                                                          |                       |
| Reserved Field                         | 147    | 8      | Date      | Reserved for future use                                                                                                                                          |                       |
| Reserved Field                         | 155    | 8      | Date      | Reserved for future use                                                                                                                                          |                       |
| Ex Marker Code                         | 163    | 2      | Alpha     | The value of an Ex-Marker pertaining to a tradable instrument.                                                                                                   |                       |
| Security Type                          | 165    | 1      | UInt8     | Type of security.<br><br>Please refer to the Additional Field Values section of this document for valid values.                                                  |                       |
| Country Of Register                    | 166    | 3      | Alpha     | Country of Register.                                                                                                                                             |                       |
| Exchange Market Size                   | 169    | 8      | UInt64    | The Exchange Market Size (EMS) is set to show the minimum size a market maker must quote in an individual security for all executable and non executable quotes. |                       |

| Field                        | Offset | Length | Type   | Description                                                                                                                                                |             |
|------------------------------|--------|--------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Minimum Peak Size Multiplier | 177    | 8      | Size   | Used to specify the minimum size of an iceberg peak for an instrument in conjunction with EMS.                                                             |             |
| Security Maximum Spread      | 185    | 8      | Price  | This field informs Participants of the maximum spread allowable for an instrument when submitting quote messages, calculated as a percentage of mid-price. |             |
| Clearing Type                | 193    | 1      | UInt8  | Indicates the settlement mode of the security:                                                                                                             |             |
|                              |        |        |        | Value                                                                                                                                                      | Meaning     |
|                              |        |        |        | 0                                                                                                                                                          | Not Cleared |
|                              |        |        |        | 1                                                                                                                                                          | Cleared     |
| Strike Price                 | 194    | 8      | Price  | Strike Price (exercise price for warrants).                                                                                                                |             |
| Security Exchange            | 202    | 11     | Alpha  | Not Applicable to LSE                                                                                                                                      |             |
| Reserved Field               | 213    | 12     | Alpha  | Reserved for future use                                                                                                                                    |             |
| Reserved Field               | 225    | 1      | UInt8  | Reserved for future use                                                                                                                                    |             |
| Reserved Field               | 226    | 8      | UInt64 | Reserved for future use                                                                                                                                    |             |
| Reserved Field               | 234    | 8      | Size   | Reserved for future use                                                                                                                                    |             |
| Partition ID                 | 242    | 1      | Byte   | Trading System's partition in which the instrument is traded                                                                                               |             |
| Reserved Field               | 243    | 8      | Size   | Reserved for future use                                                                                                                                    |             |
| Reserved Field               | 251    | 8      | Price  | Reserved for future use.                                                                                                                                   |             |
| Reserved Field               | 259    | 4      | UInt32 | Reserved for future use.                                                                                                                                   |             |
| Reserved Field               | 263    | 2      | UInt16 | Reserved for future use.                                                                                                                                   |             |
| Symbol                       | 265    | 8      | Alpha  | Symbol of the instrument.                                                                                                                                  |             |
| Description                  | 273    | 40     | Alpha  | Description of the instrument                                                                                                                              |             |

## 3.11.4 Instrument Status

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | Y           | Y           | Y                   | Y                   | Y         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field          | Offset | Length | Type   | Description                                                  |                                       |
|----------------|--------|--------|--------|--------------------------------------------------------------|---------------------------------------|
| Length         | 0      | 2      | UInt16 | Length of message including this field.                      |                                       |
| Message Type   | 2      | 1      | Byte   | <b>Hex</b>                                                   | <b>Meaning</b>                        |
|                |        |        |        | 0x48                                                         | Instrument Status                     |
| Timestamp      | 3      | 8      | UDT    | Time the message was generated.                              |                                       |
| Instrument     | 11     | 8      | UInt64 | GTP Instrument identifier.                                   |                                       |
| Source venue   | 19     | 2      | UInt16 | Venue from which market data is received for the instrument. |                                       |
|                |        |        |        | <b>Value</b>                                                 | <b>Meaning</b>                        |
|                |        |        |        | 1                                                            | London Stock Exchange                 |
| Trading Status | 21     | 1      | Byte   | <b>Value</b>                                                 | <b>Meaning</b>                        |
|                |        |        |        | H                                                            | Halt                                  |
|                |        |        |        | J                                                            | Halt – Matching partition Suspended   |
|                |        |        |        | K                                                            | Halt – System Suspended               |
|                |        |        |        | T                                                            | Regular Trading/Start Trade Reporting |
|                |        |        |        | P                                                            | Halt – Regulatory                     |
|                |        |        |        | t                                                            | End Trade Reporting                   |
|                |        |        |        | a                                                            | Opening Auction Call                  |

| Field                 | Offset | Length | Type  | Description                                                                                                                                                                                                                   |
|-----------------------|--------|--------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |        |        |       | b Post-Close                                                                                                                                                                                                                  |
|                       |        |        |       | c Closed                                                                                                                                                                                                                      |
|                       |        |        |       | d Closing Auction Call                                                                                                                                                                                                        |
|                       |        |        |       | e AESP Auction Call                                                                                                                                                                                                           |
|                       |        |        |       | f Resume Auction                                                                                                                                                                                                              |
|                       |        |        |       | m Pre-Mandatory                                                                                                                                                                                                               |
|                       |        |        |       | n Mandatory                                                                                                                                                                                                                   |
|                       |        |        |       | o Post-Mandatory                                                                                                                                                                                                              |
|                       |        |        |       | q EDSP Auction Call                                                                                                                                                                                                           |
|                       |        |        |       | r Periodic Auction Call                                                                                                                                                                                                       |
|                       |        |        |       | 1 Inactive                                                                                                                                                                                                                    |
|                       |        |        |       | 2 Suspended                                                                                                                                                                                                                   |
|                       |        |        |       | w No Active Session                                                                                                                                                                                                           |
|                       |        |        |       | x End of Post Close                                                                                                                                                                                                           |
|                       |        |        |       | u Closing Price Crossing session                                                                                                                                                                                              |
| Session Change Reason | 22     | 1      | UInt8 | <b>Value</b> <b>Meaning</b>                                                                                                                                                                                                   |
|                       |        |        |       | 0 Scheduled Transition                                                                                                                                                                                                        |
|                       |        |        |       | 1 Extended by Market Ops                                                                                                                                                                                                      |
|                       |        |        |       | 2 Shortened by Market Ops                                                                                                                                                                                                     |
|                       |        |        |       | 3 Market Order Imbalance                                                                                                                                                                                                      |
|                       |        |        |       | 4 Price Outside Range                                                                                                                                                                                                         |
|                       |        |        |       | 5 AESP/Circuit Breaker Tripped                                                                                                                                                                                                |
|                       |        |        |       | 9 Unavailable                                                                                                                                                                                                                 |
| New End Time          | 23     | 6      | Time  | New time the session will end. The field will contain only spaces if Session Change Reason is "0" or the Session Change Reason is not present. New End Time will be in terms of the local time on the server (i.e., not UTC). |
| Order Book Type       | 29     | 1      | UInt8 | <b>Value</b> <b>Name</b>                                                                                                                                                                                                      |
|                       |        |        |       | 1 Firm Quote Book                                                                                                                                                                                                             |
|                       |        |        |       | 2 Off-book                                                                                                                                                                                                                    |
|                       |        |        |       | 3 Electronic Order Book                                                                                                                                                                                                       |



## 3.11.5 Add Order – MBO

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field          | Offset | Length | Type   | Description                                                  |                       |
|----------------|--------|--------|--------|--------------------------------------------------------------|-----------------------|
| Length         | 0      | 2      | UInt16 | Length of message including this field.                      |                       |
| Message Type   | 2      | 1      | Byte   | Hex                                                          | Meaning               |
|                |        |        |        | 0x41                                                         | Add Order – MBO       |
| Timestamp      | 3      | 8      | UDT    | Time the message was generated.                              |                       |
| Order ID       | 11     | 8      | UInt64 | Unique identifier of the order.                              |                       |
| Side           | 19     | 1      | Byte   | Value                                                        | Meaning               |
|                |        |        |        | B                                                            | Buy Order             |
|                |        |        |        | S                                                            | Sell Order            |
| Size           | 20     | 8      | Size   | Displayed size of the order.                                 |                       |
| Instrument     | 28     | 8      | UInt64 | GTP Instrument identifier.                                   |                       |
| Price          | 36     | 8      | Price  | Limit price of the order                                     |                       |
| Reserved Field | 44     | 8      | Price  | Reserved for future use                                      |                       |
| Source venue   | 52     | 2      | UInt16 | Venue from which market data is received for the instrument. |                       |
|                |        |        |        | Value                                                        | Meaning               |
|                |        |        |        | 1                                                            | London Stock Exchange |

| Field           | Offset | Length | Type  | Description                                                                                                  |                       |
|-----------------|--------|--------|-------|--------------------------------------------------------------------------------------------------------------|-----------------------|
| Order Book Type | 54     | 1      | UInt8 | Value                                                                                                        | Name                  |
|                 |        |        |       | 1                                                                                                            | Firm Quote Book       |
|                 |        |        |       | 3                                                                                                            | Electronic Order Book |
| Participant     | 55     | 11     | Alpha | Identity of trading participant that submitted the order.                                                    |                       |
| Depth           | 66     | 1      | UInt8 | Total number of orders disseminated including this one, on this side of the book as indicated by Side field. |                       |

## 3.11.6 Add Order Short – MBO

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | Y           | N           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field          | Offset | Length | Type   | Description                                                |                       |
|----------------|--------|--------|--------|------------------------------------------------------------|-----------------------|
| Length         | 0      | 2      | UInt16 | Length of message including this field.                    |                       |
| Message Type   | 2      | 1      | Byte   | Hex                                                        | Meaning               |
|                |        |        |        | 0x65                                                       | Add Order Short – MBO |
| Order ID       | 3      | 8      | UInt64 | Unique identifier of the order.                            |                       |
| Size           | 11     | 8      | Size   | Displayed size of the order.                               |                       |
| Price          | 19     | 8      | Price  | Limit price of the order                                   |                       |
| Reserved Field | 27     | 8      | Price  | Reserved for future use                                    |                       |
| Participant    | 35     | 11     | Alpha  | Identity, if trading participant that submitted the order. |                       |

## 3.11.7 Add Order – MBP

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field          | Offset | Length | Type   | Description                                                  |                       |
|----------------|--------|--------|--------|--------------------------------------------------------------|-----------------------|
| Length         | 0      | 2      | UInt16 | Length of message including this field.                      |                       |
| Message Type   | 2      | 1      | Byte   | <b>Hex</b>                                                   | <b>Meaning</b>        |
|                |        |        |        | 0x66                                                         | Add Order MBP         |
| Timestamp      | 3      | 8      | UDT    | Time the message was generated.                              |                       |
| Side           | 11     | 1      | Byte   | <b>Value</b>                                                 | <b>Meaning</b>        |
|                |        |        |        | B                                                            | Buy Order             |
|                |        |        |        | S                                                            | Sell Order            |
| Size           | 12     | 8      | Size   | Displayed size of the order.                                 |                       |
| Instrument     | 20     | 8      | UInt64 | GTP Instrument identifier.                                   |                       |
| Price          | 28     | 8      | Price  | Limit price of the order                                     |                       |
| Reserved Field | 36     | 8      | Price  | Reserved for future use                                      |                       |
| Source venue   | 44     | 2      | UInt16 | Venue from which market data is received for the instrument. |                       |
|                |        |        |        | <b>Value</b>                                                 | <b>Meaning</b>        |
|                |        |        |        | 1                                                            | London Stock Exchange |

| Field           | Offset | Length | Type   | Description                                                                                                  |                       |
|-----------------|--------|--------|--------|--------------------------------------------------------------------------------------------------------------|-----------------------|
| Order Book Type | 46     | 1      | UInt8  | Value                                                                                                        | Name                  |
|                 |        |        |        | 1                                                                                                            | Firm Quote Book       |
|                 |        |        |        | 3                                                                                                            | Electronic Order Book |
| Splits          | 47     | 2      | UInt16 | Number of orders at this price point.                                                                        |                       |
| Depth           | 49     | 1      | UInt8  | Total number of orders disseminated including this one, on this side of the book as indicated by Side field. |                       |

## 3.11.8 Add Order Short – MBP

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | Y           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field          | Offset | Length | Type   | Description                                                                        |
|----------------|--------|--------|--------|------------------------------------------------------------------------------------|
| Length         | 0      | 2      | UInt16 | Length of message including this field.                                            |
| Message Type   | 2      | 1      | Byte   | <div>Hex</div> <div>Meaning</div> <div>0x67</div> <div>Add Order Short – MBP</div> |
| Size           | 3      | 8      | Size   | Displayed size of the order.                                                       |
| Price          | 11     | 8      | Price  | Limit price of the order                                                           |
| Reserved Field | 19     | 8      | Price  | Reserved for future use                                                            |
| Splits         | 27     | 2      | UInt16 | Number of orders at this price point.                                              |

## 3.11.9 Add Order Incremental

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | Y                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field            | Offset | Length | Type   | Description                                                        |                       |
|------------------|--------|--------|--------|--------------------------------------------------------------------|-----------------------|
| Length           | 0      | 2      | UInt16 | Length of message including this field.                            |                       |
| Message Type     | 2      | 1      | Byte   | <b>Hex</b>                                                         | <b>Meaning</b>        |
|                  |        |        |        | 0x46                                                               | Add Order Incremental |
| Timestamp        | 3      | 8      | UDT    | Time the message was generated.                                    |                       |
| Order ID         | 11     | 8      | UInt64 | Unique identifier of the order.                                    |                       |
| Side             | 19     | 1      | Byte   | <b>Value</b>                                                       | <b>Meaning</b>        |
|                  |        |        |        | B                                                                  | Buy Order             |
|                  |        |        |        | S                                                                  | Sell Order            |
| Size             | 20     | 8      | Size   | Displayed Size of the order.                                       |                       |
| Instrument       | 28     | 8      | UInt64 | GTP Instrument identifier.                                         |                       |
| Price            | 36     | 8      | Price  | Limit price of the order                                           |                       |
| Transaction Time | 44     | 8      | UDT    | Transaction execution timestamp as reported by the upstream system |                       |

| Field           | Offset | Length | Type   | Description                                                  |                                  |
|-----------------|--------|--------|--------|--------------------------------------------------------------|----------------------------------|
| Source venue    | 52     | 2      | UInt16 | Venue from which market data is received for the instrument. |                                  |
|                 |        |        |        | <b>Value</b>                                                 | <b>Meaning</b>                   |
|                 |        |        |        | 1                                                            | London Stock Exchange            |
| Order Book Type | 54     | 1      | UInt8  | <b>Value</b>                                                 | <b>Name</b>                      |
|                 |        |        |        | 1                                                            | Firm Quote Book                  |
|                 |        |        |        | 3                                                            | Electronic Order Book            |
|                 |        |        |        | 4                                                            | Private RFQ                      |
| Participant     | 55     | 11     | Alpha  | Identity of trading participant that submitted the order.    |                                  |
| Order Type      | 66     | 1      | UInt8  | <b>Value</b>                                                 | <b>Meaning</b>                   |
|                 |        |        |        | 0                                                            | Limit Order/Firm Quote (default) |
|                 |        |        |        | 1                                                            | Market Order                     |
|                 |        |        |        | 3                                                            | Named Limit                      |
|                 |        |        |        | 4                                                            | Named Market                     |
|                 |        |        |        | 5                                                            | Executable Quotes                |
| RFQ ID          | 67     | 10     | Alpha  | Unique ID assigned to a particular RFQ by the system.        |                                  |
|                 |        |        |        | Only relevant when Order Book Type is 4 = Private RFQ.       |                                  |



## 3.11.10 Order Modify

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field           | Offset | Length | Type      | Description                             |                       |                                          |
|-----------------|--------|--------|-----------|-----------------------------------------|-----------------------|------------------------------------------|
| Length          | 0      | 2      | UInt16    | Length of message including this field. |                       |                                          |
| Message Type    | 2      | 1      | Byte      | Hex                                     | Meaning               |                                          |
|                 |        |        |           | 0x55                                    | Modify Order          |                                          |
| Timestamp       | 3      | 8      | UDT       | Time the message was generated.         |                       |                                          |
| Order ID        | 11     | 8      | UInt64    | Unique identifier of the order.         |                       |                                          |
| Instrument      | 19     | 8      | UInt64    | GTP Instrument identifier.              |                       |                                          |
| Side            | 27     | 1      | Byte      | Value                                   | Meaning               |                                          |
|                 |        |        |           | B                                       | Buy Order             |                                          |
|                 |        |        |           | S                                       | Sell Order            |                                          |
| Flags           | 28     | 1      | Bit Field | Bit                                     | Name                  | Meaning                                  |
|                 |        |        |           | 0                                       | Priority Flag         | 0: Priority Lost<br>1: Priority Retained |
| Order Book Type | 29     | 1      | UInt8     | Value                                   | Name                  |                                          |
|                 |        |        |           | 1                                       | Firm Quote Book       |                                          |
|                 |        |        |           | 3                                       | Electronic Order Book |                                          |
|                 |        |        |           | 4                                       | Private RFQ           |                                          |
| New Quantity    | 30     | 8      | Size      | New displayed quantity of the order.    |                       |                                          |

| Field             | Offset | Length | Type   | Description                                                        |                       |
|-------------------|--------|--------|--------|--------------------------------------------------------------------|-----------------------|
| New Price         | 38     | 8      | Price  | New price of the order.                                            |                       |
| Reserved Field    | 46     | 8      | Price  | Reserved for future use                                            |                       |
| Source venue      | 54     | 2      | UInt16 | Venue from which market data is received for the instrument.       |                       |
|                   |        |        |        | Value                                                              | Meaning               |
|                   |        |        |        | 1                                                                  | London Stock Exchange |
| Previous Price    | 56     | 8      | Price  | Previous price of the order.                                       |                       |
| Previous Quantity | 64     | 8      | Size   | Previous displayed quantity of the order.                          |                       |
| Transaction Time  | 72     | 8      | UDT    | Transaction execution timestamp as reported by the upstream system |                       |

## 3.11.11 Order Delete

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field           | Offset | Length | Type   | Description                                                  |                       |
|-----------------|--------|--------|--------|--------------------------------------------------------------|-----------------------|
| Length          | 0      | 2      | UInt16 | Length of message including this field.                      |                       |
| Message Type    | 2      | 1      | Byte   | <b>Hex</b>                                                   | <b>Meaning</b>        |
|                 |        |        |        | 0x44                                                         | Delete Order          |
| Timestamp       | 3      | 8      | UDT    | Time the message was generated                               |                       |
| Order ID        | 11     | 8      | UInt64 | Unique identifier of the order.                              |                       |
| Instrument      | 19     | 8      | UInt64 | GTP Instrument identifier.                                   |                       |
| Side            | 27     | 1      | Byte   | <b>Value</b>                                                 | <b>Meaning</b>        |
|                 |        |        |        | B                                                            | Buy Order             |
|                 |        |        |        | S                                                            | Sell Order            |
| Order Book Type | 28     | 1      | UInt8  | <b>Value</b>                                                 | <b>Name</b>           |
|                 |        |        |        | 1                                                            | Firm Quote Book       |
|                 |        |        |        | 3                                                            | Electronic Order Book |
|                 |        |        |        | 4                                                            | Private RFQ           |
| Source venue    | 29     | 2      | UInt16 | Venue from which market data is received for the instrument. |                       |
|                 |        |        |        | <b>Value</b>                                                 | <b>Meaning</b>        |
|                 |        |        |        | 1                                                            | London Stock Exchange |
| Previous Price  | 31     | 8      | Price  | Price of the order that was deleted from the book.           |                       |

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| Field             | Offset | Length | Type | Description                                                        |
|-------------------|--------|--------|------|--------------------------------------------------------------------|
| Previous Quantity | 39     | 8      | Size | Quantity of the order that was deleted from the book.              |
| Transaction Time  | 47     | 8      | UDT  | Transaction execution timestamp as reported by the upstream system |

## 3.11.12 Top of Book

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | N           | N           | N                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field             | Offset | Length | Type   | Description                                                                                |                       |
|-------------------|--------|--------|--------|--------------------------------------------------------------------------------------------|-----------------------|
| Length            | 0      | 2      | UInt16 | Length of message including this field.                                                    |                       |
| Message Type      | 2      | 1      | Byte   | Hex                                                                                        | Meaning               |
|                   |        |        |        | 0x69                                                                                       | TOB                   |
| Timestamp         | 3      | 8      | UDT    | Time the message was generated.                                                            |                       |
| Instrument        | 11     | 8      | UInt64 | GTP Instrument identifier.                                                                 |                       |
| Source venue      | 19     | 2      | UInt16 | Venue from which market data is received for the instrument.                               |                       |
|                   |        |        |        | Value                                                                                      | Meaning               |
|                   |        |        |        | 1                                                                                          | London Stock Exchange |
| Bid Market Size   | 21     | 8      | Size   | Aggregated size of all bid market orders. Value will be 0 if there are no market orders.   |                       |
| Bid Limit Price   | 29     | 8      | Price  | Price of the best buy limit order                                                          |                       |
| Reserved Field    | 37     | 8      | Price  | Reserved for future use                                                                    |                       |
| Bid Limit Size    | 45     | 8      | Size   | Aggregated size of all orders at the best buy limit price.                                 |                       |
| Offer Market Size | 53     | 8      | Size   | Aggregated size of all offer market orders. Value will be 0 if there are no market orders. |                       |
| Offer Limit Price | 61     | 8      | Price  | Price of the best sell limit order                                                         |                       |
| Reserved Field    | 69     | 8      | Price  | Reserved for future use                                                                    |                       |
| Offer Limit Size  | 77     | 8      | Size   | Aggregated size of all orders at the best sell limit price.                                |                       |

| Field           | Offset | Length | Type      | Description |                       |                   |
|-----------------|--------|--------|-----------|-------------|-----------------------|-------------------|
| Order Book Type | 85     | 1      | UInt8     | Value       | Name                  |                   |
|                 |        |        |           | 1           | Firm Quote Book       |                   |
|                 |        |        |           | 3           | Electronic Order Book |                   |
| Flags           | 86     | 1      | Bit Field | Bit         | Name                  | Meaning           |
|                 |        |        |           | 0           | Bid Depth             | 0 – No<br>1 – Yes |
|                 |        |        |           | 1           | Offer Depth           | 0 – No<br>1 – Yes |

## 3.11.13 Order Book Clear

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field           | Offset | Length | Type   | Description                                                  |                       |
|-----------------|--------|--------|--------|--------------------------------------------------------------|-----------------------|
| Length          | 0      | 2      | UInt16 | Length of message including this field.                      |                       |
| Message Type    | 2      | 1      | Byte   | Hex                                                          | Meaning               |
|                 |        |        |        | 0x79                                                         | Order Book Clear      |
| Timestamp       | 3      | 8      | UDT    | Time the message was generated.                              |                       |
| Source venue    | 11     | 2      | UInt16 | Venue from which market data is received for the instrument. |                       |
|                 |        |        |        | Value                                                        | Meaning               |
|                 |        |        |        | 1                                                            | London Stock Exchange |
| Instrument      | 13     | 8      | UInt64 | GTP Instrument identifier.                                   |                       |
| Order Book Type | 21     | 1      | UInt8  | Value                                                        | Name                  |
|                 |        |        |        | 1                                                            | Firm Quote Book       |
|                 |        |        |        | 3                                                            | Electronic Order Book |
|                 |        |        |        | 4                                                            | Private RFQ           |

## 3.11.14 Trade

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | Y        |

| Field            | Offset | Length | Type   | Description                                                                                                                                                         |
|------------------|--------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length           | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                             |
| Message Type     | 2      | 1      | Byte   | <div>Hex</div> <div>Meaning</div>                                                                                                                                   |
|                  |        |        |        | 0x50 Trade                                                                                                                                                          |
| Timestamp        | 3      | 8      | UDT    | Time the message was generated.                                                                                                                                     |
| Transaction Time | 11     | 8      | UDT    | Execution timestamp as reported by the supported market.<br>If a trade is cancelled or amended, this field will contain the transaction time of the original trade. |
| Source venue     | 19     | 2      | UInt16 | Venue from which market data is received for the instrument.                                                                                                        |
|                  |        |        |        | <div>Value</div> <div>Meaning</div>                                                                                                                                 |
|                  |        |        |        | 1 London Stock Exchange                                                                                                                                             |
| Executed Size    | 21     | 8      | Size   | Total executed quantity.                                                                                                                                            |
| Instrument       | 29     | 8      | UInt64 | GTP Instrument identifier.                                                                                                                                          |
| Price            | 37     | 8      | Price  | Executed price                                                                                                                                                      |
| Reserved Field   | 45     | 8      | Price  | Reserved for future use                                                                                                                                             |
| Trade ID         | 53     | 8      | UInt64 | Unique identifier of the trade.                                                                                                                                     |



| Field                      | Offset | Length | Type      | Description                                                                                                                                                                                                                             |                                    |                 |
|----------------------------|--------|--------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------|
| Trade Type                 | 61     | 1      | UInt8     | Value                                                                                                                                                                                                                                   | Meaning                            |                 |
|                            |        |        |           | 0                                                                                                                                                                                                                                       | Regular (or Continuous) Trade      |                 |
|                            |        |        |           | 1                                                                                                                                                                                                                                       | Auction Trade – Bulk               |                 |
|                            |        |        |           | 2                                                                                                                                                                                                                                       | Auction Trade – Individual         |                 |
|                            |        |        |           | 9                                                                                                                                                                                                                                       | On-book Trade Cancellation         |                 |
|                            |        |        |           | 11                                                                                                                                                                                                                                      | Trade Correction                   |                 |
|                            |        |        |           | 22                                                                                                                                                                                                                                      | RFQ Trade                          |                 |
|                            |        |        |           | 23                                                                                                                                                                                                                                      | RFQ Trade Cancellation             |                 |
|                            |        |        |           | 24                                                                                                                                                                                                                                      | RFQ Trade Correction               |                 |
| Auction Type               | 62     | 1      | Byte      | The value in this field is only relevant when Trade Type is 1:                                                                                                                                                                          |                                    |                 |
|                            |        |        |           | Value                                                                                                                                                                                                                                   | Meaning                            |                 |
|                            |        |        |           | C                                                                                                                                                                                                                                       | Closing Auction                    |                 |
|                            |        |        |           | O                                                                                                                                                                                                                                       | Opening Auction                    |                 |
|                            |        |        |           | A                                                                                                                                                                                                                                       | AESP                               |                 |
|                            |        |        |           | B                                                                                                                                                                                                                                       | EDSP (Specific to LSE)             |                 |
|                            |        |        |           | E                                                                                                                                                                                                                                       | Resume Auction (Specific to LSE)   |                 |
|                            |        |        |           | F                                                                                                                                                                                                                                       | Periodic Auction (Specific to LSE) |                 |
| Flags                      | 63     | 1      | Bit Field | Bit                                                                                                                                                                                                                                     | Name                               | Meaning         |
|                            |        |        |           | 0                                                                                                                                                                                                                                       | Trade Cancellation                 | 0: No<br>1: Yes |
|                            |        |        |           | 1                                                                                                                                                                                                                                       | Trade Correction                   | 0: No<br>1: Yes |
| Hidden Execution Indicator | 64     | 1      | UInt8     | Value                                                                                                                                                                                                                                   | Meaning                            |                 |
|                            |        |        |           | 0                                                                                                                                                                                                                                       | N/A                                |                 |
|                            |        |        |           | 1                                                                                                                                                                                                                                       | Visible                            |                 |
|                            |        |        |           | 2                                                                                                                                                                                                                                       | Hidden                             |                 |
|                            |        |        |           | For London Stock Exchange, during continuous trading and CPX session, 1[2] will be sent for execution of visible[hidden] quantities.<br><br>For London Stock Exchange Auction Trades, 0 will be sent indicating ‘Not Applicable’ (N/A). |                                    |                 |
| Trade Qualifier            | 65     | 1      | Byte      | Value                                                                                                                                                                                                                                   | Meaning                            |                 |
|                            |        |        |           | Space                                                                                                                                                                                                                                   | N/A                                |                 |
|                            |        |        |           | C                                                                                                                                                                                                                                       | Closing Price Cross (CPX)          |                 |

## 3.11.15 Statistics

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | N        |

| Field                           | Offset | Length | Type   | Description                                                          |
|---------------------------------|--------|--------|--------|----------------------------------------------------------------------|
| Length                          | 0      | 2      | UInt16 | Length of message including this field.                              |
| Message Type                    | 2      | 1      | Byte   | <b>Hex</b> <b>Meaning</b>                                            |
|                                 |        |        |        | 0x77                      Statistics                                 |
| Timestamp                       | 3      | 8      | UDT    | Time the message was generated.                                      |
| Instrument                      | 11     | 8      | UInt64 | GTP Instrument identifier.                                           |
| Source venue                    | 19     | 2      | UInt16 | Venue from which market data is received for the instrument.         |
|                                 |        |        |        | <b>Value</b> <b>Meaning</b>                                          |
|                                 |        |        |        | 1                      London Stock Exchange                         |
| Volume                          | 21     | 8      | Size4  | Cumulative volume of all trades for the trading day.                 |
| Volume (on-book only)           | 29     | 8      | Size4  | Cumulative volume for the trading day excluding off-book trades.     |
| VWAP                            | 37     | 8      | Price4 | Volume weighted average price for the day for all trades.            |
| VWAP (on-book only)             | 45     | 8      | Price4 | Volume weighted average price for the day excluding off-book trades. |
| Number of trades                | 53     | 4      | UInt32 | Count of all trades for the day.                                     |
| Number of trades (on-book only) | 57     | 4      | UInt32 | Count of trades for the day excluding off-book trades.               |
| Turnover                        | 61     | 8      | Price4 | Turnover of all trades for the day.                                  |
| Turnover (on-book only)         | 69     | 8      | Price4 | Turnover for the day excluding off-book trades.                      |

## 3.11.16 Statistics Update

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | Y       | Y           | Y           | Y                   | N                   | N         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | Y       | N/A         | N/A         | N/A                 | N/A                 | N/A       | Y      | N        |

| Field          | Offset | Length | Type   | Description                                                    |                                    |
|----------------|--------|--------|--------|----------------------------------------------------------------|------------------------------------|
| Length         | 0      | 2      | UInt16 | Length of message including this field.                        |                                    |
| Message Type   | 2      | 1      | Byte   | Hex                                                            | Meaning                            |
|                |        |        |        | 0x6A                                                           | Statistics Update                  |
| Timestamp      | 3      | 8      | UDT    | Time the message was generated.                                |                                    |
| Instrument     | 11     | 8      | UInt64 | GTP Instrument identifier.                                     |                                    |
| Source venue   | 19     | 2      | UInt16 | Venue from which market data is received for the instrument.   |                                    |
|                |        |        |        | Value                                                          | Meaning                            |
|                |        |        |        | 1                                                              | London Stock Exchange              |
| Statistic Type | 21     | 2      | UInt16 | The statistic that is disseminated with this message instance: |                                    |
|                |        |        |        | Value                                                          | Meaning                            |
|                |        |        |        | 1                                                              | Indicative Auction Uncrossing Data |
|                |        |        |        | 2                                                              | Official Opening Price             |
|                |        |        |        | 3                                                              | Official Closing Price             |
|                |        |        |        | 4                                                              | Trade High On-Book                 |
|                |        |        |        | 5                                                              | Trade Low On-Book                  |
|                |        |        |        | 6                                                              | Trade High All Trades              |
|                |        |        |        | 7                                                              | Trade Low All Trades               |

| Field                           | Offset | Length | Type  | Description                                                                                                                  |
|---------------------------------|--------|--------|-------|------------------------------------------------------------------------------------------------------------------------------|
|                                 |        |        |       | 8 52-week Trade High All Trades                                                                                              |
|                                 |        |        |       | 9 52-week Trade Low All Trades                                                                                               |
|                                 |        |        |       | 10 Best Closing Bid                                                                                                          |
|                                 |        |        |       | 11 Best Closing Ask                                                                                                          |
|                                 |        |        |       | 16 Static Reference Price                                                                                                    |
|                                 |        |        |       | 17 Dynamic Reference Price                                                                                                   |
| Statistic Price                 | 23     | 8      | Price | The value of price type statistics<br>If the Opening or Closing Price is cleared manually by the venue, '-1' will be stamped |
| Statistic Size                  | 31     | 8      | Size  | The value of size type statistics.                                                                                           |
| Auction Type                    | 39     | 1      | Byte  | Populated if the Statistic Type is 1:                                                                                        |
|                                 |        |        |       | <b>Value</b> <b>Meaning</b>                                                                                                  |
|                                 |        |        |       | C Closing Auction                                                                                                            |
|                                 |        |        |       | O Opening Auction                                                                                                            |
|                                 |        |        |       | A AESP                                                                                                                       |
|                                 |        |        |       | B EDSP                                                                                                                       |
|                                 |        |        |       | E Resume Auction                                                                                                             |
|                                 |        |        |       | F Periodic Auction                                                                                                           |
| Imbalance Quantity              | 40     | 8      | Size  | Quantity that is eligible to be matched at the indicative price but will not be matched.                                     |
| Auction Info                    | 48     | 1      | Byte  | Populated if the Statistic Type is 1:                                                                                        |
|                                 |        |        |       | <b>Value</b> <b>Meaning</b>                                                                                                  |
|                                 |        |        |       | 0 Not Applicable                                                                                                             |
|                                 |        |        |       | B Buy Imbalance                                                                                                              |
|                                 |        |        |       | N No Imbalance                                                                                                               |
|                                 |        |        |       | O Insufficient Orders for Auction                                                                                            |
|                                 |        |        |       | S Sell Imbalance                                                                                                             |
| Opening/Closing Price Indicator | 49     | 1      | Byte  | Populated if the Statistic Type is 2 or 3:                                                                                   |
|                                 |        |        |       | <b>Value</b> <b>Meaning</b>                                                                                                  |
|                                 |        |        |       | A UT                                                                                                                         |
|                                 |        |        |       | B AT                                                                                                                         |
|                                 |        |        |       | C Mid of BBO                                                                                                                 |
|                                 |        |        |       | D Last AT                                                                                                                    |
|                                 |        |        |       | E Last UT                                                                                                                    |
|                                 |        |        |       | F Manual                                                                                                                     |
|                                 |        |        |       | I Previous Close                                                                                                             |

## 3.11.17 Statistics Snapshot

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | N         | N      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | Y        |

| Field                           | Offset | Length | Type   | Description                                                          |                       |
|---------------------------------|--------|--------|--------|----------------------------------------------------------------------|-----------------------|
| Length                          | 0      | 2      | UInt16 | Length of message including this field.                              |                       |
| Message Type                    | 2      | 1      | Byte   | Hex                                                                  | Meaning               |
|                                 |        |        |        | 0x6b                                                                 | Statistics Snapshot   |
| Timestamp                       | 3      | 8      | UDT    | Time the message was generated.                                      |                       |
| Instrument                      | 11     | 8      | UInt64 | GTP Instrument identifier.                                           |                       |
| Source venue                    | 19     | 2      | UInt16 | Venue from which market data is received for the instrument.         |                       |
|                                 |        |        |        | Value                                                                | Meaning               |
|                                 |        |        |        | 1                                                                    | London Stock Exchange |
| Volume                          | 21     | 8      | Size4  | Cumulative volume of all trades for the trading day.                 |                       |
| Volume (on-book only)           | 29     | 8      | Size4  | Cumulative volume for the trading day excluding off-book trades.     |                       |
| VWAP                            | 37     | 8      | Price4 | Volume weighted average price for the day for all trades.            |                       |
| VWAP (on-book only)             | 45     | 8      | Price4 | Volume weighted average price for the day excluding off-book trades. |                       |
| Number of trades                | 53     | 4      | UInt32 | Count of all trades for the day.                                     |                       |
| Number of trades (on-book only) | 57     | 4      | UInt32 | Count of trades for the day excluding off-book trades.               |                       |
| Turnover                        | 61     | 8      | Price4 | Turnover of all trades for the day.                                  |                       |

| Field                     | Offset | Length | Type   | Description                                                                                                                                            |
|---------------------------|--------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Turnover (on-book only)   | 69     | 8      | Price4 | Turnover for the day excluding off-book trades.                                                                                                        |
| Official Opening Price    | 77     | 8      | Price  | Official Opening Price for the instrument.<br>If the Opening Price is cleared manually by the venue '-1' will be stamped.                              |
| Official Closing Price    | 85     | 8      | Price  | Official Closing Price for the instrument.<br>If the Closing Price is cleared manually by the venue '-1' will be stamped.                              |
| Trade High (on-book only) | 93     | 8      | Price  | Current trading day high price excluding off-book trades.                                                                                              |
| Trade Low (on-book only)  | 101    | 8      | Price  | Current trading day low price excluding off-book trades.                                                                                               |
| Trade High                | 109    | 8      | Price  | Current trading day high price of all trades.                                                                                                          |
| Trade Low                 | 117    | 8      | Price  | Current trading day low price of all trades.                                                                                                           |
| 52-week Trade High        | 125    | 8      | Price  | 52-week high price of all trades.                                                                                                                      |
| 52-week Trade Low         | 133    | 8      | Price  | 52-week low price of all trades.                                                                                                                       |
| Opening Price Indicator   | 141    | 1      | Byte   | Please refer to Description in Statistics Update message for valid values.<br>This will be blank if no Opening Price is contained within the snapshot. |
| Closing Price Indicator   | 142    | 1      | Byte   | Please refer to Description in Statistics Update message for valid values.<br>This will be blank if no Closing Price is contained within the snapshot. |
| IAU Price                 | 143    | 8      | Price  | Contains the last reported Indicative Auction Crossing Price                                                                                           |
| IAU Paired Size           | 151    | 8      | Size   | Quantity to be matched at the last reported indicative price.                                                                                          |
| Imbalance Quantity        | 159    | 8      | Size   | Quantity that was eligible to be matched at the indicative price but was not matched at the last indicative price.                                     |
| Imbalance Direction       | 167    | 1      | Byte   | <b>Value</b> <b>Meaning</b>                                                                                                                            |
|                           |        |        |        | B      Buy Imbalance                                                                                                                                   |
|                           |        |        |        | N      No Imbalance                                                                                                                                    |
|                           |        |        |        | O      Insufficient Orders for Auction                                                                                                                 |
|                           |        |        |        | S      Sell Imbalance                                                                                                                                  |
| Best Closing Bid Price    | 168    | 8      | Price  | The best bid price at the time the instrument moves into closing auction session.                                                                      |
| Best Closing Ask Price    | 176    | 8      | Price  | The best offer price at the time the instrument moves into closing auction session.                                                                    |
| Best Closing Bid Size     | 184    | 8      | Size   | The best bid size at the time the instrument moves into closing auction session.                                                                       |
| Best Closing Ask Size     | 192    | 8      | Size   | The best offer size at the time the instrument moves into closing auction session.                                                                     |
| Reserved Field            | 200    | 8      | Price  | Reserved for future use                                                                                                                                |

| Field                   | Offset | Length | Type  | Description                                                                                                                              |                  |
|-------------------------|--------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Reserved Field          | 208    | 8      | Price | Reserved for future use                                                                                                                  |                  |
| Reserved Field          | 216    | 8      | Size  | Reserved for future use                                                                                                                  |                  |
| Reserved Field          | 224    | 8      | Price | Reserved for future use                                                                                                                  |                  |
| Auction Type            | 232    | 1      | Byte  | <b>Value</b>                                                                                                                             | <b>Meaning</b>   |
|                         |        |        |       | C                                                                                                                                        | Closing Auction  |
|                         |        |        |       | O                                                                                                                                        | Opening Auction  |
|                         |        |        |       | A                                                                                                                                        | AESP             |
|                         |        |        |       | B                                                                                                                                        | EDSP             |
|                         |        |        |       | E                                                                                                                                        | Resume Auction   |
|                         |        |        |       | F                                                                                                                                        | Periodic Auction |
| Last Trade Price        | 233    | 8      | Price | Executed price at which the instrument was last traded. If no relevant trades have taken place the value '0' will be populated.          |                  |
| Last Trade Quantity     | 241    | 8      | Size  | Executed quantity of the trade which set the 'Last Trade Price'. If no relevant trades have taken place the value '0' will be populated. |                  |
| Last Trade Time         | 249    | 8      | UDT   | Transaction time of the trade which set the 'Last Trade Price'. If no relevant trades have taken place the value '0' will be populated.  |                  |
| Static Reference Price  | 257    | 8      | Price | Reference Price as reported by the source venue.                                                                                         |                  |
| Dynamic Reference Price | 265    | 8      | Price | Reference Price as reported by the source venue.                                                                                         |                  |

## 3.11.18 MiFID II Trade

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | Y                   | N         | Y      | Y        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field                           | Offset | Length | Type   | Description                                                  |                               |
|---------------------------------|--------|--------|--------|--------------------------------------------------------------|-------------------------------|
| Length                          | 0      | 2      | UInt16 | Length of message including this field.                      |                               |
| Message Type                    | 2      | 1      | Byte   | Hex                                                          | Meaning                       |
|                                 |        |        |        | 0x51                                                         | MiFID Trade                   |
| Timestamp                       | 3      | 8      | UDT    | Message dissemination time.                                  |                               |
| Source venue                    | 11     | 2      | UInt16 | Venue from which market data is received for the instrument. |                               |
|                                 |        |        |        | Value                                                        | Meaning                       |
|                                 |        |        |        | 1                                                            | London Stock Exchange         |
| Instrument                      | 13     | 8      | UInt64 | GTP Instrument identifier.                                   |                               |
| Transaction identification code | 21     | 52     | Alpha  | A unique trade identifier. The value will be right aligned.  |                               |
| Trade Type                      | 73     | 1      | UInt8  | Value                                                        | Meaning                       |
|                                 |        |        |        | 0                                                            | Regular (or Continuous) Trade |
|                                 |        |        |        | 1                                                            | Auction Trade – Bulk          |
|                                 |        |        |        | 2                                                            | Auction Trade – Individual    |
|                                 |        |        |        | 9                                                            | Trade Cancellation            |
|                                 |        |        |        | 11                                                           | Trade Correction              |
|                                 |        |        |        | 22                                                           | RFQ Trade                     |
|                                 |        |        |        | 23                                                           | RFQ Trade Cancellation        |



| Field                               | Offset | Length | Type          | Description                                                                                                                                                                                                                                      |
|-------------------------------------|--------|--------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Auction Type                        | 74     | 1      | Byte          | 24 RFQ Trade Correction                                                                                                                                                                                                                          |
|                                     |        |        |               | The value in this field is only relevant when Trade Type is 1.                                                                                                                                                                                   |
|                                     |        |        |               | <b>Value</b> <b>Meaning</b>                                                                                                                                                                                                                      |
|                                     |        |        |               | C Closing Auction                                                                                                                                                                                                                                |
|                                     |        |        |               | O Opening Auction                                                                                                                                                                                                                                |
|                                     |        |        |               | A AESP                                                                                                                                                                                                                                           |
|                                     |        |        |               | B EDSP (Specific to LSE)                                                                                                                                                                                                                         |
|                                     |        |        |               | E Resume Auction                                                                                                                                                                                                                                 |
| MiFID Price                         | 75     | 20     | MiFID Decimal | F Periodic Auction                                                                                                                                                                                                                               |
|                                     |        |        |               | MiFID compliant Price field populated using either Price or Percentage.                                                                                                                                                                          |
|                                     |        |        |               | {DECIMAL-18/13} in case the price is expressed as monetary value.<br>{DECIMAL-11/10} in case the price is expressed as percentage                                                                                                                |
| MiFID Quantity                      | 95     | 20     | MiFID Decimal | Number of units of the financial instrument. {DECIMAL-18/17}<br><b>Will be set to default value (20 spaces) if the traded security is a bond</b>                                                                                                 |
| Trading Date and Time               | 115    | 27     | Date and Time | Date and time when the transaction was executed/agreed upon.<br>If a trade is cancelled or amended, this field will contain the Trading Date and Time of the original trade.                                                                     |
| Instrument Identification Code Type | 142    | 4      | Alpha         | <b>Value</b> <b>Meaning</b>                                                                                                                                                                                                                      |
|                                     |        |        |               | ISIN International Securities Identification Number                                                                                                                                                                                              |
|                                     |        |        |               | Only relevant for non-equity instruments.<br>Empty for equity and equity-like instruments.                                                                                                                                                       |
| Instrument Identification Code      | 146    | 12     | Alpha         | Instrument identification number (ISIN code).                                                                                                                                                                                                    |
| Price Notation                      | 158    | 4      | Alpha         | Indicates if the price is expressed in monetary value or in percentage.<br>Possible values:                                                                                                                                                      |
|                                     |        |        |               | <b>Value</b> <b>Meaning</b>                                                                                                                                                                                                                      |
|                                     |        |        |               | MONE Monetary value                                                                                                                                                                                                                              |
|                                     |        |        |               | PERC Percentage                                                                                                                                                                                                                                  |
|                                     |        |        |               | Only relevant for non-equity instruments.                                                                                                                                                                                                        |
| Price Major Currency                | 162    | 3      | Alpha         | Major currency in which the price is expressed (applicable if the price is expressed as monetary value).<br>Currency Code as per ISO 4217.<br>For additional currencies supported refer to the Additional Field Values section of this document. |
| Notional Amount                     | 165    | 20     | MiFID Decimal | Notional value relevant to the security. {DECIMAL-18/5}<br>Only relevant for non-equity instruments.                                                                                                                                             |

| Field                     | Offset | Length | Type          | Description                                                                                                                              |
|---------------------------|--------|--------|---------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Notional Currency         | 185    | 3      | Alpha         | Major Currency in which the notional amount is denominated.<br>Only relevant for non-equity instruments.                                 |
| Venue of Execution        | 188    | 4      | Alpha         | Identification of the venue where the transaction was executed.                                                                          |
| Publication Date and Time | 192    | 27     | Date and Time | Date and time when the transaction was published.                                                                                        |
| PTRefPriceWaiverFlag      | 219    | 4      | Alpha         | Not Applicable to LSE                                                                                                                    |
| Reserved Field            | 223    | 4      | Alpha         | Reserved for future use                                                                                                                  |
| Market Closing Price Flag | 227    | 4      | Alpha         | Possible values:<br><b>Value</b><br>CLSE                                                                                                 |
| PTAlgoTrade               | 231    | 4      | Alpha         | Possible values:<br><b>Value</b><br>ALGO <sup>1</sup>                                                                                    |
| PTCancellationFlag        | 235    | 4      | Alpha         | Possible values:<br><b>Value</b><br>CANC                                                                                                 |
| PTAmendmentFlag           | 239    | 4      | Alpha         | Possible values:<br><b>Value</b><br>AMND                                                                                                 |
| Reserved Field            | 243    | 1      | Byte          | Reserved for future use                                                                                                                  |
| Reserved Field            | 244    | 3      | Alpha         | Reserved for future use                                                                                                                  |
| Reserved Field            | 247    | 20     | MiFID Decimal | Reserved for future use                                                                                                                  |
| Reserved Field            | 267    | 4      | Alpha         | Reserved for future use                                                                                                                  |
| Trade Qualifier           | 271    | 1      | Byte          | <b>Value</b> <b>Meaning</b><br>Space      N/A<br>C      Closing Price Cross (CPX)                                                        |
| Market Mechanism          | 272    | 1      | Byte          | <b>Value</b> <b>Meaning</b><br>1      Central Limit Order Book<br>6      Request for Quote                                               |
| Trading Mode              | 273    | 1      | Byte          | <b>Value</b> <b>Meaning</b><br>O      Scheduled Opening Auction<br>K      Scheduled Closing Auction<br>I      Scheduled Intraday Auction |

<sup>1</sup> ALGO flag not disseminated on Auction Bulk Trades

| Field                        | Offset | Length | Type | Description  |                                            |
|------------------------------|--------|--------|------|--------------|--------------------------------------------|
|                              |        |        |      | U            | Unscheduled Auction                        |
|                              |        |        |      | 2            | Continuous Trading                         |
|                              |        |        |      | 3            | At Market Close Trading                    |
| Transaction Category         | 274    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | D            | Dark Trade <sup>2</sup>                    |
|                              |        |        |      | -            | None apply                                 |
| Negotiation Indicator        | 275    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | -            | Not a Negotiated Trade                     |
| Agency Cross Indicator       | 276    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | -            | No Agency Cross Trade                      |
| Modification Indicator       | 277    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | C            | Trade Cancellation (CANC)                  |
|                              |        |        |      | A            | Trade Amendment (AMND)                     |
|                              |        |        |      | -            | New Trade                                  |
| Reference Price Indicator    | 278    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | 1            | Market Closing Price Trade (CLSE)          |
|                              |        |        |      | -            | Not a Reference Price Trade                |
| Special Dividend Indicator   | 279    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | -            | No Special Dividend Trade                  |
| Off Book Automated Indicator | 280    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | -            | Unspecified or does not apply              |
| Price Formation Indicator    | 281    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | P            | Plain-Vanilla Trade                        |
| Algorithmic Indicator        | 282    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | H            | Algorithmic Trade (ALGO)                   |
|                              |        |        |      | -            | Not an Algorithmic Trade                   |
| Post-Trade Deferral Reason   | 283    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | -            | Immediate Publication                      |
| Deferral/ Enrichment Type    | 284    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | -            | Not Applicable/No Relevant Enrichment Type |
| Duplicative Indicator        | 285    | 1      | Byte | <b>Value</b> | <b>Meaning</b>                             |
|                              |        |        |      | -            | Unique Trade Report                        |

<sup>2</sup> Dark Trades will not be flagged during Auctions on London Stock Exchange

### 3.11.19 Trade Summary

Trade Summary message is used to disseminate results of multi- and single-fill trade events.

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | Y                   | N                   | N         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field                          | Offset | Length | Type   | Description                                                                                                                                                                                                                                                                                    |
|--------------------------------|--------|--------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length                         | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                                                                                                                                                        |
| Message Type                   | 2      | 1      | Byte   | <b>Hex</b> <b>Meaning</b><br>0x57      Trade Summary                                                                                                                                                                                                                                           |
| Timestamp                      | 3      | 8      | UDT    | Time the message was generated.                                                                                                                                                                                                                                                                |
| Instrument                     | 11     | 8      | UInt64 | GTP Instrument identifier.                                                                                                                                                                                                                                                                     |
| Source Venue                   | 19     | 2      | UInt16 | Venue from which market data is received for the instrument.<br><b>Value</b> <b>Meaning</b><br>1      London Stock Exchange                                                                                                                                                                    |
| Transaction Time               | 21     | 8      | UDT    | Execution timestamp reported by the supported Source Venue.                                                                                                                                                                                                                                    |
| Far Price                      | 29     | 8      | Price  | Far price is the highest price at which volume is depleted as part of the Matching Engine event in case of an aggressing buy order, or the lowest price at which volume is depleted in case of an aggressing sell order.<br>Determined by considering all trades in one Matching Engine event. |
| Total Executed Quantity        | 37     | 8      | Size   | Total quantity executed (visible & hidden quantity).                                                                                                                                                                                                                                           |
| Total Hidden Executed Quantity | 45     | 8      | Size   | Total hidden order quantity executed (hidden quantity).                                                                                                                                                                                                                                        |

| Field                  | Offset | Length | Type     | Description                                                                                                                                                                                                     |                                         |
|------------------------|--------|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Deleted Order Quantity | 53     | 8      | Size     | Order Quantity deleted due to Self Execution Prevention (SEP) during an execution.                                                                                                                              |                                         |
| Side                   | 61     | 1      | Byte     | The side of the visible passive order(s) that triggered the trade summary.                                                                                                                                      |                                         |
|                        |        |        |          | Value                                                                                                                                                                                                           | Meaning                                 |
|                        |        |        |          | Space                                                                                                                                                                                                           | No Side (only Hidden quantity executed) |
|                        |        |        |          | B                                                                                                                                                                                                               | Buy Side                                |
|                        |        |        |          | S                                                                                                                                                                                                               | Sell Side                               |
| Best Bid Size          | 62     | 8      | Int Size | Aggregated size of all orders at the best buy price available on the book at execution completion<br>Value will be set to -1 if top of the book is not disclosed                                                |                                         |
| Best Bid Price         | 70     | 8      | Price    | Price of best visible bid available on the book at execution completion<br>Value will be set to -1 if top of the book is not disclosed<br>Value will be set to 0 in case of market order is on top of the book  |                                         |
| Best Offer Size        | 78     | 8      | Int Size | Aggregated size of all orders at the best sell price available on the book at execution completion<br>Value will be set to -1 if top of the book is not disclosed                                               |                                         |
| Best Offer Price       | 86     | 8      | Price    | Price of best visible bid available on the book at execution completion<br>Value will be set to -1 if top of the book is not disclosed.<br>Value will be set to 0 in case of market order is on top of the book |                                         |

### 3.11.20 Analytics

Analytics Message is used to disseminate additional statistics including order book activity statistics.

| Market Data Group | Source Venue          | Channel Content                          | Level 1 | Level 2 MBO | Level 2 MBP | Level 2 Incremental | MIFID II Post Trade | Analytics | Replay | Recovery |
|-------------------|-----------------------|------------------------------------------|---------|-------------|-------------|---------------------|---------------------|-----------|--------|----------|
| Channel A/a       | London Stock Exchange | FTSE 100                                 | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel B/b       | London Stock Exchange | ETFs / ETCs / ETNs - 1                   | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel C/c       | London Stock Exchange | Other Main Market Equity & Miscellaneous | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel D/d       | London Stock Exchange | ETFs / ETCs / ETNs - 2                   | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel E/e       | London Stock Exchange | AIM                                      | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel F/f       | London Stock Exchange | European                                 | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel G/g       | London Stock Exchange | International                            | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel H/h       | London Stock Exchange | Fixed Income and Securitised Derivatives | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel J/j       | London Stock Exchange | ETFs / ETCs / ETNs - 3                   | N       | N           | N           | N                   | N                   | Y         | Y      | N        |
| Channel P/p       | London Stock Exchange | Private Securities Market                | N       | N/A         | N/A         | N/A                 | N/A                 | N/A       | N      | N        |

| Field            | Offset | Length | Type   | Description                                                                                                                                               |
|------------------|--------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length           | 0      | 2      | UInt16 | Length of message including this field.                                                                                                                   |
| Message Type     | 2      | 1      | Byte   | <div>Hex</div> <div>Meaning</div> <div>0x61</div> <div>Analytics</div>                                                                                    |
| Timestamp        | 3      | 8      | UDT    | Time the message was generated.                                                                                                                           |
| Instrument       | 11     | 8      | UInt64 | GTP Instrument identifier.                                                                                                                                |
| Source Venue     | 19     | 2      | UInt16 | <div>Venue from which market data is received for the instrument.</div> <div>Value</div> <div>Meaning</div> <div>1</div> <div>London Stock Exchange</div> |
| Start Time       | 21     | 8      | UDT    | Time the calculation of the statistics on this message began.                                                                                             |
| End Time         | 29     | 8      | UDT    | Time the calculation of the statistics on this message ended.                                                                                             |
| Buy Order Count  | 37     | 4      | UInt32 | Number of buy orders received within the calculation window.                                                                                              |
| Sell Order Count | 41     | 4      | UInt32 | Number of sell orders received within the calculation window.                                                                                             |
| Buy Order Size   | 45     | 8      | Size4  | Cumulative quantity of all buy orders received within the calculation window.                                                                             |
| Sell Order Size  | 53     | 8      | Size4  | Cumulative quantity of all sell orders received within the calculation window.                                                                            |

| Field                           | Offset | Length | Type   | Description                                                                                                                                                                                                                                                                                     |
|---------------------------------|--------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Buy Order Cancellations         | 61     | 4      | UInt32 | Number of buy orders cancelled by clients within the calculation window.                                                                                                                                                                                                                        |
| Sell Order Cancellations        | 65     | 4      | UInt32 | Number of sell orders cancelled by clients within the calculation window.                                                                                                                                                                                                                       |
| Buy Limit Order Cancellations   | 69     | 4      | UInt32 | Number of buy limit orders cancelled by clients within the calculation window.                                                                                                                                                                                                                  |
| Buy Market Order Cancellations  | 73     | 4      | UInt32 | Number of buy market orders cancelled by clients within the calculation window.                                                                                                                                                                                                                 |
| Sell Limit Order Cancellations  | 77     | 4      | UInt32 | Number of sell limit orders cancelled by clients within the calculation window.                                                                                                                                                                                                                 |
| Sell Market Order Cancellations | 81     | 4      | UInt32 | Number of sell market orders cancelled by clients within the calculation window.                                                                                                                                                                                                                |
| Bid Ask Spread                  | 85     | 8      | Price  | <p>Most Recent Bid/Ask spread at the time of publication of the message.</p> <p>If the value is set to zero, this means, either:</p> <ol style="list-style-type: none"> <li>1) No sell/buy liquidity</li> <li>2) Order book is locked or crossed</li> <li>3) No orders on order book</li> </ol> |
| VWAP Buy                        | 93     | 8      | Price  | Volume Weighted Average Price for trades triggered by an aggressing buy order. Calculated within the calculation window for trades executed in continuous trading.                                                                                                                              |
| VWAP Sell                       | 101    | 8      | Price  | Volume Weighted Average Price for trades triggered by an aggressing sell order. Calculated within the calculation window for trades executed in continuous trading.                                                                                                                             |

## 4 Client Data Recovery

The Group Ticker Plant operates with similar data recovery solutions as the existing Millennium Exchange market data product: replay and recovery service.

Should a gap in sequence numbers be noticed on both the primary and secondary feed, or following a failure at client site, clients should assume that instrument order books are stale and should initiate one of the below data recovery processes to refresh systems.

### 4.1 Replay Channel

The replay service provides clients the ability to request a finite number of application messages as disseminated on the real-time multicast channel. Provisioned to facilitate client recovery following a small scale data loss, the replay service caches the last 65,000 application messages published for each multicast channel.

Clients are permitted to log on to the replay service a finite number of times each day and, following successful login, subsequently submit a finite number of requests each day. Whilst these counters can be reset intraday by the Group, this will be done only in an emergency situation and should not be relied upon as normal practice.

Each client CompID is permitted to log on to the Production replay service 3,000 times per day, submitting a maximum of 3,000 requests per day. Clients are permitted to queue a maximum of 10 requests at any one time. Should any of these parameters be reached, the replay server will respond with an explanatory reason code.

#### 4.1.1 Establishing a connection

Clients should send a Login Request message to the appropriate target replay service IP gateway address and port. Validation of appropriate credentials as configured on the Group Ticker Plant will be done against both the CompID, as supplied in the message sent by the client, and the incoming source IP address. Upon successful validation, the replay service responds with a Login Response message of Status 'A' – the session should now be considered active. Clients should wait for the server's response prior to submitting replay requests. Any requests submitted prior to authentication will be ignored.

Should the attempted connection fail for any reason, the server will respond with a Login Response message which will contain a reason code. This reason code will facilitate the diagnosis of the failed login request.

#### 4.1.2 Sending a request

Once successful connection is established, clients can queue requests for retransmission of missed messages. The Replay Request message should include the first sequence number of the range of messages to be retransmitted and the total number of messages subsequent to the first missed message required. A Request ID can also be included if required – this is not validated by the replay service.

Should the server accept the request for message retransmission, clients will receive a Replay Response message of Status 'A.' If a Request ID was specified, this will be included in the Replay Response message. This will be immediately followed by the requested message stream. The completion of the request will be marked by the sending of a Replay and Recovery Complete message. This will include the original Request ID if specified but will not include the current Trading Status of the instrument.

Should the replay request fail, the server will respond with a Replay Response message which will contain a reason code. This reason code will facilitate the diagnosis of the failed replay request.

Submitted requests will be processed serially, but the capability of the replay service will be split across any logged in CompID. This may mean that the performance of the replay solution may differ, dependent upon the number of CompIDs logged in to the service at any given time.

#### 4.1.3 Terminating a connection

Clients will not be required to log out from the replay service. Instead, immediately after the completion of the request, the replay server will terminate the connection with the client.

Clients should note that, upon successful login to the replay service, a request should be submitted within five seconds or the server will force-logout the client.

Replay, Recovery and GTP Lite services will close connection to user via message with FIN flag in case termination of connection is initiated by the service itself.



## 4.2 Recovery Channel

The recovery service facilitates a client's resynchronisation with the order book following a large scale data loss for the which the replay service is insufficient. Following successful login, clients are able to request the following:

- A snapshot of the order book for any active instrument in the Market Data Group
- All trades reported for the trading day (both on and off book)
- A snapshot of an instrument's current statistics
- The current trading status of an instrument
- The full set of reference data for an instrument

In a similar approach to that of the replay service, clients are permitted to log in to the recovery service a finite number of times each day and, following successful login, subsequently submit a finite number of requests each day. While these counters can be reset intraday by the Group, this will be done only in an emergency and should not be relied upon as normal practice. Each client CompID is permitted to log on to the Production recovery service 3,000 times per day, submitting a maximum of 3,000 requests per day. Clients are permitted to queue a maximum of 10 requests at any one time. Should any of these parameters be reached, the recovery server will respond with an explanatory reason code.

### 4.2.1 Establishing a connection

Clients should send a Login Request message to the appropriate target recovery service IP gateway address and port. Validation of appropriate credentials as configured on the Group Ticker Plant will be done against both the CompID as supplied in the message by the client and the incoming source IP address. Upon successful validation, the recovery service will respond with a Login Response message of Status 'A' – the session should now be considered active. Clients should wait for the server's response prior to submitting recovery requests. Any requests submitted prior to authentication will be ignored.

Should the attempted connection fail for any reason, the server will respond with a Login Response message which will contain a reason code. This reason code will facilitate the diagnosis of the failed login request.

### 4.2.2 Sending a request for an Instrument Level Order Book snapshot

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for a snapshot of the current order book using the Recovery Request message. The Recovery Request message should indicate that the client requires an instrument order book snapshot – the Recovery Type field should be '1,' and the Request Level field should be '0.' Clients are also permitted to request a specific order book, either Electronic or Firm Quote, in the Order Book Type field. Clients may also include a Request ID in the Recovery Request message – this is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will also include the Request ID if specified by the client. The Recovery Response message will also include the real-time channel sequence number with which the snapshot is synchronised. Clients should buffer all messages as received on the real-time channel with a sequence number greater than that received in the Recovery Response message.

If active and displayable orders are present on the order book, the server will transmit an Order Book Clear message followed by a message stream to allow the rebuild of the order book. The MBO and MBP snapshot service line order book recovery will be disseminated through a number of Add Order messages. The order book should be built in the same way as on the real-time channel – further details are contained within section 3.1.2 of GTP001 – Product Guide. The level 2 incremental services should be rebuilt by processing all Add Order Incremental messages as disseminated following the Recovery Response message. The buy side will always be transmitted first. The level 1 order book recovery will publish an Order Book Clear message followed by the current best Bid and Ask aggregated levels in a single TOB message, similar to that of the real-time service.

If the snapshot request was for more than one order book as specified in the Order Book Type field of the Recovery Request message, the Electronic Order Book will always be provided first. Following transmission of each requested order book, the server will disseminate a Replay and Recovery Complete message. If no active displayable orders are present on any applicable order book type, the server will disseminate only an Order Book Clear message followed by the Replay and Recovery Complete message. This message includes the real-time trading status of the instrument if the instrument is currently in a trading session. To complete the request, following transmission of all requested order books for an instrument, the server will disseminate a final Replay and Recovery Complete message. If a Request ID was specified by the client, this will be included.

### 4.2.3 Sending a request for a Group or Channel Level Order Book snapshot

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for a snapshot of the current order book using the Recovery Request message. The Recovery Request message should indicate that the client requires Group or Channel level order book snapshots – the Recovery Type field should be '1,' and the Request Level field should be either '1' or '2'. If a Group or Channel level recovery is requested by the client, all order books will be disseminated – the Order Book Type field is not processed by the server. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will also include the Request ID, if specified by the client, and the real-time channel sequence number with which the snapshot is synchronised. Clients should buffer all messages as received on the real-time channel with a sequence number greater than that received in the Recovery Response message.

If active and displayable orders are present on the order book, the server will transmit an Order Book Clear message, followed by a message stream to allow the rebuild of the order book. The MBO and MBP snapshot service line order book recovery will be disseminated through a number of Add Order and Add Order Short messages. The order book should be built in the same way as on the real-time channel – further details are contained within section 3.1.2 of GTP001 – Product Guide.

The level 2 incremental services should be rebuilt by processing all Add Order Incremental messages as disseminated following the Recovery Response message. The buy side will always be transmitted first. Order books will be transmitted to clients serially, with the Electronic Order Book always transmitted prior to the Firm Quote book if applicable. If private RFQ book is applicable, a series of Add Order Incremental messages or a series of Indicative Quote Information messages will be transmitted pertaining to private RFQ book depending on the RFQ Transparency regime. All applicable order books will be transmitted prior to dissemination of the next instrument's snapshot. The level 1 order book recovery will publish an Order Book Clear message followed by the current best Bid and Ask aggregated levels in a single TOB message – similar to that of the real-time service.

For instruments with no active displayable orders present on any applicable order book type, the server will disseminate only an Order Book Clear message followed by the Replay and Recovery Complete message. This message includes the real-time trading status of the instrument if the instrument is currently in a trading session. For the private RFQ book, the Trading Status will always be set to No Active Session (w). To complete the full request, following transmission of all requested order books, the server will disseminate a final Replay and Recovery Complete message. If a Request ID was specified by the client, this will be included. The final Replay and Recovery Complete message will not contain a value in the Trading Status field.

### 4.2.4 Sending a request for the Instrument Directory (reference data)

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for Instrument Directory messages using the Recovery Request message. The Recovery Request message should indicate that the client requires Instrument Directory messages. The Recovery Type field should be '0.' Clients should also indicate the level of the request – an individual instrument, a segment or for all instruments on the multicast channel. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will also include the Request ID, if specified by the client. The Recovery Response message will also include the real-time channel sequence number of the last Instrument Directory message sent. This will be followed by the Instrument Directory message(s) as requested by the client. Successful 'Group' or 'Multicast Channel' requests will result in the dissemination of all configured instruments on the Group Ticker Plant at that request level, irrespective of their trading status. Instrument Directory messages are available via Recovery once they have been published by multicast.

Customers should note that, should a request be made for an instrument not supported by the targeted multicast group, the server will respond with the rejection Status of 'a.'

The completion of the recovery request will be indicated through the dissemination of a Replay and Recovery Complete message. The Trading Status field of the Replay and Recovery Complete message will only be populated if the original request level was 'Instrument.' The Replay and Recovery Complete message will also include the Request ID, if specified by the client.

#### 4.2.5 Sending a request for trades

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for trades as reported by supported markets using the Recovery Request message. The Recovery Request message should indicate that the client requires trade recovery. The Recovery Type field should be '2.' Clients should also indicate the level of the request – an individual instrument, a segment or for all instruments on the multicast channel. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message, which should indicate the successful acceptance of the request. The Recovery Response message will include the Request ID, as specified by the client. This message will also include the real-time multicast channel sequence number of the last trade to be disseminated as part of the request, with the total number of trade messages to be disseminated indicated in the Count field.

The Recovery Response message will be immediately followed by a stream of execution messages as disseminated on the multicast channel. All trade types will be disseminated, including Trade, Off-book Trade, and Trade Cross messages, in their original sequence. It is not possible to request a subset of trade types on the trade recovery service. Trade cancellations as originally disseminated will be included in the recovery service message stream. While clients cannot specify a subset of trade types on the recovery service, clients may include a real-time channel sequence number on the Recovery Request message. When a sequence number is included, the recovery service will transmit only trade messages with an equal or greater sequence number to that specified.

The completion of the recovery request will be indicated through the dissemination of a Replay and Recovery Complete message. The Trading Status field of the Replay and Recovery Complete message will only be populated if the original request level was 'Instrument.' The Replay and Recovery Complete message will also include the Request ID, if specified by the client.

If no trade messages exist which satisfy the original request, the server will transmit a Recovery Response message followed immediately by a Replay and Recovery Complete message.

#### 4.2.6 Sending a request for a statistics snapshot

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for an instrument's statistics as calculated by the Group Ticker Plant by using the Recovery Request message. The Recovery Request message should indicate that the client requires statistics. The Recovery Type field should be '3.' Clients should also indicate the level of the request – an individual instrument, a segment or for all instruments on the multicast channel. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will include the Request ID, as specified by the client. This message will also include the real-time multicast channel sequence number of the last message sent on the multicast channel. Clients should buffer all messages as received on the real-time channel with a sequence number greater than specified. The total number of Statistic Snapshot messages to be disseminated will be indicated in the Count field.

Following dissemination of the Recovery Response message, the server will disseminate one or more Statistic Snapshot messages, the number dependent upon the level of the original request. The Statistic Snapshot message will provide clients all current statistics as calculated by the Group Ticker Plant.

The server will disseminate a Replay and Recovery Complete message to indicate the successful completion of the request. The Trading Status field will only be populated if the request level was 'Instrument'. The Replay and Recovery Complete message will also include the Request ID if specified by the client.

If no statistics exist to satisfy the original request, the server will transmit a Recovery Response message followed immediately by a Replay and Recovery Complete message.

#### 4.2.8 Sending a request for an Instrument Status

Following receipt of a Login Response message of Status 'A,' used to confirm successful login to the recovery service, clients may submit a request for the current Trading Status of an instrument on any supported market by using the Recovery Request message. The Recovery Request message should indicate that the client requires Instrument Status messages. The Recovery Type field should be '4.' Clients should also indicate the level of the request – an individual instrument, a segment or for all instruments on the multicast channel. Clients may also include a Request ID in the Recovery Request message. This is not validated by the server.

The server will transmit a Recovery Response message which should indicate the successful acceptance of the request. The Recovery Response message will include the Request ID, as specified by the client. This message will also include the real-time multicast channel sequence number of the last real-time message disseminated. Clients should buffer all messages as received on the real-time channel with a sequence number greater than specified. The total number of Instrument Status messages to be disseminated will be indicated in the Count field.

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Following dissemination of the Recovery Response message, the server will transmit one or more Instrument Status messages, the number dependent upon the level of the original request. The current Trading Status of an instrument is indicated in the Instrument Status message.

The server will disseminate a Replay and Recovery Complete message to indicate the successful completion of the request. The Trading Status field will only be populated if the request level was 'Instrument'. The Replay and Recovery Complete message will also include the Request ID if specified by the client.

If no Instrument Status messages exist to satisfy the original request, the server will transmit a Recovery Response message followed immediately by a Replay and Recovery Complete message.

#### **4.2.9 Terminating a connection**

Clients will not be required to log out from the recovery service. Instead, immediately after the completion of the request, the recovery server will terminate the connection with the client.

Clients should note that, upon successful login to the recovery service, a request should be submitted within five seconds or the server will force-logout the client.

Replay, Recovery and GTP Lite services will close connection to user via message with FIN flag in case termination of connection is initiated by the service itself.

## 5 Additional field values

### 5.1 Security Types

| Value | Code | Source Venue          | Description              | Security Definition |
|-------|------|-----------------------|--------------------------|---------------------|
| 1     | IE   | London Stock Exchange | International Equity     | Non-bond            |
| 4     | CN   | London Stock Exchange | Convertible Bond         | Bond                |
| 5     | RT   | London Stock Exchange | Right                    | Non-bond            |
| 39    | BD   | London Stock Exchange | Bonds                    | Bond                |
| 43    | CW   | London Stock Exchange | Covered Warrants         | Non-bond            |
| 44    | DB   | London Stock Exchange | Debenture                | Bond                |
| 45    | DE   | London Stock Exchange | UK Equity                | Non-bond            |
| 46    | DR   | London Stock Exchange | Deposit Receipts         | Non-bond            |
| 47    | EW   | London Stock Exchange | Equity Warrants          | Non-bond            |
| 48    | FB   | London Stock Exchange | Foreign Government Bonds | Bond                |
| 53    | GT   | London Stock Exchange | Gilts                    | Bond                |
| 56    | LS   | London Stock Exchange | Loan Stock               | Bond                |
| 57    | ML   | London Stock Exchange | Medium Term Loans        | Bond                |
| 64    | PR   | London Stock Exchange | Preference Shares        | Non-bond            |
| 66    | PU   | London Stock Exchange | Package Units            | Non-bond            |
| 69    | SP   | London Stock Exchange | Structured Products      | Non-bond            |

### 5.2 Supported currencies

| Code | Description   |
|------|---------------|
| EUX  | Euro – cents  |
| USX  | Cent          |
| GBX  | GB pennies    |
| ZAC  | 100th of RAND |
| ITL  | Italian LIRA  |

## 6.3 Tick ID

| GTP Tick ID | TableID | MinValue | MaxValue   | TickValue |
|-------------|---------|----------|------------|-----------|
| 1           | TM_1    | 0.0001   | 1          | 0.0001    |
|             |         | 1        | 5          | 0.0005    |
|             |         | 5        | 10         | 0.001     |
|             |         | 10       | 50         | 0.005     |
|             |         | 50       | 100        | 0.01      |
|             |         | 100      | 500        | 0.05      |
|             |         | 500      | 1,000      | 0.1       |
|             |         | 1,000    | 5,000      | 0.5       |
|             |         | 5,000    | 10,000     | 1         |
|             |         | 10,000   | 99,999,999 | 5         |
|             |         |          |            |           |
| 2           | TM_2    | 0.0001   | 0.5        | 0.0001    |
|             |         | 0.5      | 1          | 0.0005    |
|             |         | 1        | 5          | 0.001     |
|             |         | 5        | 10         | 0.005     |
|             |         | 10       | 50         | 0.01      |
|             |         | 50       | 100        | 0.05      |
|             |         | 100      | 500        | 0.1       |
|             |         | 500      | 1,000      | 0.5       |
|             |         | 1,000    | 5,000      | 1         |
|             |         | 5,000    | 10,000     | 5         |
|             |         | 10,000   | 99,999,999 | 10        |
| 3           | TM_3    | 0.0001   | 0.1        | 0.0001    |
|             |         | 0.1      | 5          | 0.0025    |
|             |         | 5        | 10         | 0.005     |
|             |         | 10       | 99,999,999 | 0.01      |
| 4           | TM_4    | 0.0001   | 0.1        | 0.0001    |
|             |         | 0.1      | 1          | 0.001     |
|             |         | 1        | 10         | 0.01      |
|             |         | 10       | 500        | 0.25      |
|             |         | 500      | 1,000      | 0.5       |
|             |         | 1,000    | 99,999,999 | 1         |
| 5           | TM_5    | 0.5      | 99,999,999 | 0.5       |
| 6           | TM_6    | 0.01     | 99,999,999 | 0.01      |
| 7           | TM_7    | 0.005    | 99,999,999 | 0.005     |
| 8           | TM_8    | 0.0025   | 99,999,999 | 0.0025    |
| 9           | TM_9    | 0.25     | 99,999,999 | 0.25      |
| 10          | TM_10   | 0.05     | 99,999,999 | 0.05      |
| 11          | TM_11   | 1        | 99,999,999 | 1         |
| 12          | TM_12   | 0.1      | 99,999,999 | 0.1       |

| GTP Tick ID | TableID | MinValue   | MaxValue   | TickValue  |
|-------------|---------|------------|------------|------------|
| 13          | TM_13   | 0.0005     | 99,999,999 | 0.0005     |
| 14          | TM_14   | 0.0001     | 99,999,999 | 0.0001     |
| 15          | TM_15   | 50         | 99,999,999 | 50         |
| 16          | TM_16   | 10         | 99,999,999 | 10         |
| 17          | TM_17   | 5          | 99,999,999 | 5          |
| 18          | TM_18   | 0.125      | 99,999,999 | 0.125      |
| 19          | TM_19   | 0.0625     | 99,999,999 | 0.0625     |
| 20          | TM_20   | 0.015625   | 99,999,999 | 0.015625   |
| 21          | TM_21   | 0.0078125  | 99,999,999 | 0.0078125  |
| 22          | TM_22   | 0.00390625 | 99,999,999 | 0.00390625 |
| 23          | TM_23   | 0.001      | 99,999,999 | 0.001      |
| 24          | TM_24   | 0.00000001 | 99,999,999 | 0.00000001 |
| 25          | TM_25   | 0.03125    | 99,999,999 | 0.03125    |
| 26          | TM_26   | 0.0001     | 10         | 0.0001     |
|             |         | 10         | 100        | 0.01       |
|             |         | 100        | 99,999,999 | 0.25       |
| 27          | TM_27   | 0.0001     | 10         | 0.0001     |
|             |         | 10         | 99,999,999 | 0.25       |
| 28          | TM_28   | 0.0001     | 10         | 0.0001     |
|             |         | 10         | 99,999,999 | 0.25       |
| 29          | TM_29   | 0.0025     | 1          | 0.0025     |
|             |         | 1          | 50         | 0.01       |
|             |         | 50         | 100        | 0.1        |
|             |         | 100        | 500        | 0.25       |
|             |         | 500        | 1,000      | 0.5        |
|             |         | 1,000      | 99,999,999 | 1          |
| 31          | TM_31   | 0.0001     | 0.1        | 0.0001     |
|             |         | 0.1        | 5          | 0.0025     |
|             |         | 5          | 10         | 0.005      |
|             |         | 10         | 99,999,999 | 0.01       |
| 32          | TM_32   | 0.0001     | 0.1        | 0.0001     |
|             |         | 0.1        | 1          | 0.001      |
|             |         | 1          | 10         | 0.01       |
|             |         | 10         | 500        | 0.25       |
|             |         | 500        | 1,000      | 0.5        |
|             |         | 1,000      | 99,999,999 | 1          |
| 33          | TM_33   | 0.0025     | 5          | 0.0025     |
|             |         | 5          | 10         | 0.005      |
|             |         | 10         | 99,999,999 | 0.01       |

| GTP Tick ID | TableID | MinValue | MaxValue    | TickValue |
|-------------|---------|----------|-------------|-----------|
| 34          | TM_34   | 0.005    | 10          | 0.005     |
|             |         | 10       | 100         | 0.01      |
|             |         | 100      | 500         | 0.25      |
|             |         | 500      | 1,000       | 0.5       |
|             |         | 1,000    | 99,999,999  | 1         |
| 35          | TM_35   | 0.0001   | 0.5         | 0.0001    |
|             |         | 0.5      | 1           | 0.0005    |
|             |         | 1        | 5           | 0.001     |
|             |         | 5        | 10          | 0.005     |
|             |         | 10       | 50          | 0.01      |
|             |         | 50       | 1,000       | 0.05      |
|             |         | 1,000    | 99,999,999  | 0.1       |
| 36          | TM_36   | 0.05     | 10          | 0.05      |
|             |         | 10       | 500         | 0.1       |
|             |         | 500      | 1,000       | 0.25      |
|             |         | 1,000    | 2,500       | 0.5       |
|             |         | 2,500    | 99,999,999  | 1         |
| 37          | TM_37   | 0.0005   | 0.1         | 0.0005    |
|             |         | 0.1      | 5           | 0.001     |
|             |         | 5        | 10          | 0.0025    |
|             |         | 10       | 25          | 0.005     |
|             |         | 25       | 99,999,999  | 0.01      |
| 38          | TM_38   | 0.001    | 10          | 0.001     |
|             |         | 10       | 50          | 0.005     |
|             |         | 50       | 100         | 0.01      |
|             |         | 100      | 999,999,999 | 0.05      |
| 41          | TM_41   | 0.01     | 10          | 0.01      |
|             |         | 10       | 50          | 0.02      |
|             |         | 50       | 100         | 0.05      |
|             |         | 100      | 99,999,999  | 0.1       |
| 42          | TM_42   | 0.0005   | 0.5         | 0.0005    |
|             |         | 0.5      | 1           | 0.001     |
|             |         | 1        | 5           | 0.005     |
|             |         | 5        | 10          | 0.01      |
|             |         | 10       | 50          | 0.05      |
|             |         | 50       | 100         | 0.1       |
|             |         | 100      | 500         | 0.5       |
|             |         | 500      | 1,000       | 1         |
|             |         | 1,000    | 5,000       | 5         |
|             |         | 5,000    | 10,000      | 10        |
|             |         | 10,000   | 99,999,999  | 50        |



| GTP Tick ID | TableID | MinValue | MaxValue   | TickValue |
|-------------|---------|----------|------------|-----------|
| 43          | TM_43   | 0.0005   | 0.1        | 0.0005    |
|             |         | 0.1      | 0.2        | 0.001     |
|             |         | 0.2      | 0.5        | 0.002     |
|             |         | 0.5      | 1          | 0.005     |
|             |         | 1        | 2          | 0.01      |
|             |         | 2        | 5          | 0.02      |
|             |         | 5        | 10         | 0.05      |
|             |         | 10       | 20         | 0.1       |
|             |         | 20       | 50         | 0.2       |
|             |         | 50       | 100        | 0.5       |
|             |         | 100      | 200        | 1         |
|             |         | 200      | 500        | 2         |
|             |         | 500      | 1,000      | 5         |
|             |         | 1,000    | 2,000      | 10        |
|             |         | 2,000    | 5,000      | 20        |
|             |         | 5,000    | 10,000     | 50        |
|             |         | 10,000   | 20,000     | 100       |
|             |         | 20,000   | 50,000     | 200       |
|             |         | 50,000   | 99,999,999 | 500       |
| 44          | TM_44   | 0.0002   | 0.1        | 0.0002    |
|             |         | 0.1      | 0.2        | 0.0005    |
|             |         | 0.2      | 0.5        | 0.001     |
|             |         | 0.5      | 1          | 0.002     |
|             |         | 1        | 2          | 0.005     |
|             |         | 2        | 5          | 0.01      |
|             |         | 5        | 10         | 0.02      |
|             |         | 10       | 20         | 0.05      |
|             |         | 20       | 50         | 0.1       |
|             |         | 50       | 100        | 0.2       |
|             |         | 100      | 200        | 0.5       |
|             |         | 200      | 500        | 1         |
|             |         | 500      | 1,000      | 2         |
|             |         | 1,000    | 2,000      | 5         |
|             |         | 2,000    | 5,000      | 10        |
|             |         | 5,000    | 10,000     | 20        |
|             |         | 10,000   | 20,000     | 50        |
|             |         | 20,000   | 50,000     | 100       |
|             |         | 50,000   | 99,999,999 | 200       |
| 45          | TM_45   | 0.0001   | 0.1        | 0.0001    |
|             |         | 0.1      | 0.2        | 0.0002    |
|             |         | 0.2      | 0.5        | 0.0005    |

| GTP Tick ID | TableID | MinValue | MaxValue   | TickValue |
|-------------|---------|----------|------------|-----------|
|             |         | 0.5      | 1          | 0.001     |
|             |         | 1        | 2          | 0.002     |
|             |         | 2        | 5          | 0.005     |
|             |         | 5        | 10         | 0.01      |
|             |         | 10       | 20         | 0.02      |
|             |         | 20       | 50         | 0.05      |
|             |         | 50       | 100        | 0.1       |
|             |         | 100      | 200        | 0.2       |
|             |         | 200      | 500        | 0.5       |
|             |         | 500      | 1,000      | 1         |
|             |         | 1,000    | 2,000      | 2         |
|             |         | 2,000    | 5,000      | 5         |
|             |         | 5,000    | 10,000     | 10        |
|             |         | 10,000   | 20,000     | 20        |
|             |         | 20,000   | 50,000     | 50        |
|             |         | 50,000   | 99,999,999 | 100       |
| 46          | TM_46   | 0.0001   | 0.1        | 0.0001    |
|             |         | 0.1      | 0.2        | 0.0001    |
|             |         | 0.2      | 0.5        | 0.0002    |
|             |         | 0.5      | 1          | 0.0005    |
|             |         | 1        | 2          | 0.001     |
|             |         | 2        | 5          | 0.002     |
|             |         | 5        | 10         | 0.005     |
|             |         | 10       | 20         | 0.01      |
|             |         | 20       | 50         | 0.02      |
|             |         | 50       | 100        | 0.05      |
|             |         | 100      | 200        | 0.1       |
|             |         | 200      | 500        | 0.2       |
|             |         | 500      | 1,000      | 0.5       |
|             |         | 1,000    | 2,000      | 1         |
|             |         | 2,000    | 5,000      | 2         |
|             |         | 5,000    | 10,000     | 5         |
|             |         | 10,000   | 20,000     | 10        |
|             |         | 20,000   | 50,000     | 20        |
|             |         | 50,000   | 99,999,999 | 50        |
| 47          | TM_47   | 0.0001   | 0.1        | 0.0001    |
|             |         | 0.1      | 0.2        | 0.0001    |
|             |         | 0.2      | 0.5        | 0.0001    |
|             |         | 0.5      | 1          | 0.0002    |
|             |         | 1        | 2          | 0.0005    |
|             |         | 2        | 5          | 0.001     |

| GTP Tick ID | TableID | MinValue | MaxValue   | TickValue |
|-------------|---------|----------|------------|-----------|
|             |         | 5        | 10         | 0.002     |
|             |         | 10       | 20         | 0.005     |
|             |         | 20       | 50         | 0.01      |
|             |         | 50       | 100        | 0.02      |
|             |         | 100      | 200        | 0.05      |
|             |         | 200      | 500        | 0.1       |
|             |         | 500      | 1,000      | 0.2       |
|             |         | 1,000    | 2,000      | 0.5       |
|             |         | 2,000    | 5,000      | 1         |
|             |         | 5,000    | 10,000     | 2         |
|             |         | 10,000   | 20,000     | 5         |
|             |         | 20,000   | 50,000     | 10        |
|             |         | 50,000   | 99,999,999 | 20        |
| 48          | TM_48   | 0.0001   | 0.1        | 0.0001    |
|             |         | 0.1      | 0.2        | 0.0001    |
|             |         | 0.2      | 0.5        | 0.0001    |
|             |         | 0.5      | 1          | 0.0001    |
|             |         | 1        | 2          | 0.0002    |
|             |         | 2        | 5          | 0.0005    |
|             |         | 5        | 10         | 0.001     |
|             |         | 10       | 20         | 0.002     |
|             |         | 20       | 50         | 0.005     |
|             |         | 50       | 100        | 0.01      |
|             |         | 100      | 200        | 0.02      |
|             |         | 200      | 500        | 0.05      |
|             |         | 500      | 1,000      | 0.1       |
|             |         | 1,000    | 2,000      | 0.2       |
|             |         | 2,000    | 5,000      | 0.5       |
|             |         | 5,000    | 10,000     | 1         |
|             |         | 10,000   | 20,000     | 2         |
|             |         | 20,000   | 50,000     | 5         |
|             |         | 50,000   | 99,999,999 | 10        |
|             |         |          |            |           |
| 49          | TM_49   | 0        | 2          | 1         |

## 6 Data Mapping

### 6.1 Conversion of Order ID

| Venue                 | Representation                                       |
|-----------------------|------------------------------------------------------|
| London Stock Exchange | Base 62 decoded value from source venue ASCII format |

### 6.2 Conversion of Trade ID

| Venue                 | Representation                                       |
|-----------------------|------------------------------------------------------|
| London Stock Exchange | Base 36 decoded value from source venue ASCII format |

### 6.3 Base36, Base62, Base64 decoding alphabets

#### 6.3.1 Base 36 with G Offset decoding alphabet

| DEC | Base 36 | DEC | Base 36 |
|-----|---------|-----|---------|
| 0   | G       | 18  | Y       |
| 1   | H       | 19  | Z       |
| 2   | I       | 20  | 0       |
| 3   | J       | 21  | 1       |
| 4   | K       | 22  | 2       |
| 5   | L       | 23  | 3       |
| 6   | M       | 24  | 4       |
| 7   | N       | 25  | 5       |
| 8   | O       | 26  | 6       |
| 9   | P       | 27  | 7       |
| 10  | Q       | 28  | 8       |
| 11  | R       | 29  | 9       |
| 12  | S       | 30  | A       |
| 13  | T       | 31  | B       |
| 14  | U       | 32  | C       |
| 15  | V       | 33  | D       |
| 16  | W       | 34  | E       |
| 17  | X       | 35  | F       |

6.3.2 Base 62

| DEC | Base 62 | DEC | Base 62 | DEC | Base 62 | DEC | Base 62 |
|-----|---------|-----|---------|-----|---------|-----|---------|
| 0   | 0       | 18  | I       | 36  | a       | 54  | s       |
| 1   | 1       | 19  | J       | 37  | b       | 55  | t       |
| 2   | 2       | 20  | K       | 38  | c       | 56  | u       |
| 3   | 3       | 21  | L       | 39  | d       | 57  | v       |
| 4   | 4       | 22  | M       | 40  | e       | 58  | w       |
| 5   | 5       | 23  | N       | 41  | f       | 59  | x       |
| 6   | 6       | 24  | O       | 42  | g       | 60  | y       |
| 7   | 7       | 25  | P       | 43  | h       | 61  | z       |
| 8   | 8       | 26  | Q       | 44  | i       |     |         |
| 9   | 9       | 27  | R       | 45  | j       |     |         |
| 10  | A       | 28  | S       | 46  | k       |     |         |
| 11  | B       | 29  | T       | 47  | l       |     |         |
| 12  | C       | 30  | U       | 48  | m       |     |         |
| 13  | D       | 31  | V       | 49  | n       |     |         |
| 14  | E       | 32  | W       | 50  | o       |     |         |
| 15  | F       | 33  | X       | 51  | p       |     |         |
| 16  | G       | 34  | Y       | 52  | q       |     |         |
| 17  | H       | 35  | Z       | 53  | r       |     |         |

## 6.4 Conversion of negative values in price fields

Signed data types, such as Price and Price4, might indicate negative values where applicable. Negative numbers are represented using the “Sign and magnitude” method. Encoding and decoding examples are provided in the following sections.

## 6.5 Encoding negative values in price fields

Decimal value = -1

Decimal value with eight implied decimal places = -100000000

Remove sign bit = 100000000

Convert to binary = 00000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Add sign bit = 10000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Hex value = 80 00 00 00 05 f5 e1 00

Hex value converted to Little endian = 00 e1 f5 05 00 00 00 80

## 6.6 Decoding negative values in price fields

Received bytes in hex = 00 e1 f5 05 00 00 00 80

Change the byte order to big endian = 80 00 00 00 05 f5 e1 00

Convert to binary = 10000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Most significant bit is set. Therefore this is a negative value.

Remove the sign bit = 00000000 00000000 00000000 00000000 00000101 11110101 11100001 00000000

Convert to decimal = 100000000

Add sign to decimal = -100000000

Mark eight implied decimal places -1.00000000